CAMBRIDGE INSTITUTE OF TECHNOLOGY BENGALURU

B.E.

In

Computer Science & Engineering

Syllabus

2023-24 Admissions

Version History
UG (CSE) Syllabus 2023-24 Admissions

Sl.No	Document Versions	Date	Remarks
1	Version 1.0	12-06-2024	3rd to 8th Semester Scheme for Batch 2023-24 presented to BoS on 12-06-2024. 3rd and 4th Semester Syllabi is presented to BoS on 12-06-2024 for the Batch 2023-24
2	Version 2.0	10-05-2025	Ratification of Scheme from 5th to 8th semester for the batch 2023-24 5th to 7th Semester Syllabi for Batch 2023-24 presented to BoS on 10-05-2025
3	Version 2.1	03-07-2025	Ratification of Scheme from 5th to 8th semester for the batch 2023-24
4	Version 3.0	29-11-2025	The 6th semester scheme has been revised wherein one PCC and its corresponding PCCL have been converted to an IPCC to align with the NEP guidelines. The same will be placed for ratification in the BoS meeting on 29-11-2025
5	Version 3.1	25-12-2025	In 6th semester, machine learning course is modified as suggested by BoS members. In accordance with the suggestion of the Principal, the Course Outcomes of the 6th semester have been generalized. As per the suggestions received, commercial links have been removed.

BoS Coordinators	Prof. Shilpa S B	
BoS Chairman	Dr. Shreekanth M Prabhu, HoD CSE	

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Course Title	Machine Learning	**Semester**	VI
Course Code	CS351	**CIE**	50
Total No. of Contact Hours	25 Th+20Tu=45 hrs	**SEE**	50
No. of Contact Hours/week (L:T:P)	2:2:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview:
The course introduces some of the basic concepts of machine learning from a mathematically well motivated perspective. It covers the primary approaches to machine learning. Several key algorithms, example data-sets and project-oriented activities are included.

Course objective is to:

- Familiarize with the basics of Machine Learning and Machine Learning process.
- Learn the basics of supervised learning algorithms - similarity based learning, regression analysis and Decision Tree.
- Understand the working of traditional machine learning approaches - Bayesian learning and the naïve Bayes algorithm, support vector machines, Artificial Neural Networks and clustering algorithms**.**

Prerequisite Knowledge: This course requires that the students are familiar with Data Structures, Discrete Mathematics and Probability.

Module-1: The Fundamentals of Machine Learning	**9 Hours**

The Fundamentals of Machine Learning:

The Machine Learning Landscape, What is Machine Learning? Why Use Machine Learning?, Types of Machine Learning Systems, Main Challenges of Machine Learning, Testing and Validating, Working with Real Data, Look at the Big Picture.

Basics of Learning Theory: Induction Biases, Modelling in machine learning

Tutorial: Data Preprocessing – Missing Data Analysis, Removal of Noisy or Outlier Data, Data Integration and Data Transformation, Feature Engineering and Dimensionality reduction.

Self-learning Activity Questions:

- What type of Machine Learning algorithm would you use to allow a robot to walk in various unknown terrains?
- What type of algorithm would you use to segment your customers into multiple groups?
- Would you frame the problem of spam detection as a supervised learning problem or an unsupervised learning problem?
- If your model performs great on the training data but generalizes poorly to new instances, what is happening? Can you name three possible solutions?

Textbook 1: Chapter 1
Textbook 2: Chapter 2.3.2, Chapter 2.10.1 to 2.10.3, Chapter 3.5,3.6.

Module-2: Similarity Based Learning and Regression Analysis	9 Hours

Similarity Based Learning:

Introduction to Similarity or Instance-based Learning; Nearest-Neighbor Learning; Weighted *K*-Nearest-Neighbor Algorithm; Nearest Centroid Classifier; Locally Weighted Regression (LWR).

Regression Analysis:

Introduction to Regression, Introduction to Linearity, Correlation, and Causation, Introduction to Linear Regression, Logistic Regression

Tutorial: Computational Tracing of Nearest-Neighbor Algorithm and Weighted *K*-Nearest-Neighbor Algorithm; Validation of Regression Methods

Self-learning Activity:

1. Load the Iris dataset.
   - Split the dataset into two separate dataframes – training and test.
   - Create the K-Nearest Neighbor model for different values of k (k=1 to 20).
2. Load the 'cars' dataset.
   - Split the dataset into two separate dataframes – training and test.
   - Apply the linear regression algorithm to the training dataset and find the equation to the linear regression line. Interpret the results. Test if your linear model has a good fit.
   - Predict for the test dataset. Visualize the training set results and test set results.

Textbook 2: Chapter 4, Chapter 5.1 to 5.4, 5.7

Module-3: Classification Models	9 Hours

Decision Tree learning:

Introduction to Decision Tree Learning Model, Decision Tree Induction Algorithms - ID3 Tree Construction, C4.5 Construction, Validating and Pruning of Decision Trees.

Bayesian Learning:

Introduction to Probability-based Learning, Fundamentals of Bayes theorem, Classification using Bayes Model, Navie Bayes algorithm, Navie Bayes algorithm for continuous attributes.

Support Vector Machines: Introduction to SVM, Optimal Hyperplane

Tutorial: Computational tracing of Navie Bayes algorithm and Navie Bayes algorithm for continuous attributes.

Self-learning Activity:

1. Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use EnjoySport data set for building the decision tree and apply this knowledge to classify a new sample.
2. Load the **Iris** dataset.
   - Split the dataset into two separate dataframes – training and test.
   - Implement the naïve Bayesian classifier for the training data set.
   - Compute the accuracy of the classifier.

Textbook 2: Chapter 6.1, 6.2.1, 6.2.2, 6.3 Chapter 8.1 to 8.4, Chapter 11.1, 11.2

Module-4: End-to-End Machine Learning Project	9 Hours

Working with Real Data, Look at the Big Picture, Get the Data, Explore and Visualize the Data to Gain Insights, Prepare the Data for Machine Learning Algorithms, Select and Train a Model, Fine-Tune Your Model.

Tutorial: Univariate Data Analysis and Visualization, Bivariate Data and Multivariate Data Analysis.

Self-learning Activity:

Use housing dataset. Try a Support Vector Machine regressor (sklearn.svm.SVR), with various hyperparameters such as kernel="linear" (with various values for the C hyperparameter) or kernel="rbf" (with various values for the C and gamma hyperparameters). Don't worry about what these hyperparameters mean for now. How does the best SVR predictor perform?

Try replacing GridSearchCV with RandomizedSearchCV.

Textbook 1: Chapter 2

Textbook 2: Chapter 2.5, 2.6

Module-5: Artificial Neural Network and Clustering	**9 Hours**

Artificial Neural Network

Introduction, Biological Neurons, Artificial Neurons, Perceptron and Learning Theory, Types of Artificial Neural Networks, Learning in a Multi-Layer Perceptron, Popular Applications of Artificial Neural Networks, Advantages and Disadvantages of ANN, Challenges of Artificial Neural Networks.

Clustering Algorithms:

Introduction to Clustering Approaches, Hierarchical Clustering Algorithms, Partitional Clustering Algorithm, Density-based Methods, Grid-based Approach, Cluster Evaluation Methods.

Tutorial: Computational tracing of Perceptron Algorithm, Multi-layer perceptron and k-means algorithm; Proximity Measures.

Self-learning Activity:

1. Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets (Titanic dataset).
2. Load the Iris dataset.
   - Apply k-Means algorithm to cluster the iris dataset.
   - Comment on the quality of clustering.
   - Plot the clusters formed.

Textbook 2: Chapter 10.1 to 10.6, 10.9 to 10.11, Chapter 13.1 to 13.6, 13.8

Follow-on Courses: Deep Learning, Natural Language Processing, Gen-AI.

Suggested Learning Resources:

Textbooks:

1	Aurelien Geron, Hands-On Machine Learning with Scikit-Learn and TensorFlow- Concepts, Tools and Techniques to Build Intelligent Systems, Third Edition, O'Reilly, 2023.
2	S. Sridhar, M Vijayalakshmi, "Machine Learning", 1st edition, Oxford, 2021.

Reference Books:

1	Tom Mitchell, Machine Learning, McGraw Hill Education (India) Edition, 2013.
2	Ethem Alpaydın, Introduction to Machine Learning, Second Edition, MIT press, 2010.
3	Srinvivasa K G and Shreedhar, "Artificial Intelligence and Machine Learning", Cengage, 2010.

Web links and Video Lectures (e-Resources):

1. https://nptel.ac.in/courses/106105152
2. https://nptel.ac.in/courses/106106139

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning

Outlier detection in Banking and insurance transaction for identifying fraudulent behaviour etc. Uncertainty and reasoning Problem- reliability of sensor used to detect pedestrians using Bayes Rule

Course Outcomes: At the end of this course, Students will be able to

CO1	Apply machine learning algorithms to create ML models.
CO2	Identify appropriate machine learning algorithms for a given task
CO3	Develop machine learning models and evaluate them using performance evaluation measures
CO4	Design a machine learning application using latest machine learning tools.

CO-PO Mapping

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	3	-
CO3	-	-	1	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	3	-	-	-	-	-	-	-	-	-

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Machine Learning Laboratory	**Semester**	VI
Course Code	CS352	**CIE Marks**	50
L:T:P	0:0:2	**SEE Marks**	50
Hrs. / Week	2	**Total Marks**	100
Credits	1	**Exam Duration**	3 Hours
Teaching Department	CS		

Course objectives: This laboratory course enables students to	
1.	Understand the importance of supervised machine Learning algorithms.
2.	Understand the importance of unsupervised machine Learning algorithms.
3.	Understand the cross-validation techniques of machine Learning algorithms.
4.	Evaluate the performance metrices of a model.

List of Experiments / Programs	**Hours**
Program 1 **Title:** Implement Linear Regression model for diabetes dataset. **Problem Description:** Linear Regression is typically used for predicting continuous variables. Diabetes dataset contains a continuous target variable (diabetes progression). Load the diabetes dataset, preprocess the data, train the model and test the model. **Method:** The method utilized here is supervised machine learning algorithm called Linear Regression for prediction of diabetes. Use the trained model to make predictions on the test set and evaluate the model's performance. **Theory Reference:** Module 2	2
Program 2 **Title:** Implement Logistic Regression model for Breast cancer dataset. **Problem Description:** Logistic regression is commonly used for binary classification tasks, where the goal is to predict one of two possible outcomes. Breast cancer dataset contains information about breast cancer tumors, with features like radius, texture, smoothness, and compactness. The target variable is binary, where 1 means malignant and 0 means benign. Load the dataset, preprocess the data, train the model and make predictions **Method:** The method utilized here is supervised machine learning algorithm called Logistic Regression for prediction of breast cancer from the dataset. Use the trained model to make predictions on the test set and evaluate accuracy, confusion matrix, and classification report. **Theory Reference:** Module 2	2

Program 3 **Title:** Evaluation of Regression Model Using MAE, MSE, RMSE, and MAPE **Problem Description:** 1. Load the Boston Housing dataset. 2. Split the dataset. 3. Train the model 4. Make predictions using the trained model. 5. Evaluate the model using the common evaluation metrics 6. **MAE**: We use mean_absolute_error() to calculate the mean absolute error. 7. **MSE**: We use mean_squared_error() to calculate the mean squared error. 8. **RMSE**: We compute RMSE by taking the square root of MSE. 9. **MAPE**: We manually calculate MAPE using the formula provided, taking the absolute percentage errors for each prediction. **Method:** These regression metrics (MAE, MSE, RMSE, and MAPE) gives the performance of the model from different perspectives. **Theory Reference:** Module 3	2
Program 4: **Title**: Write a program to construct the Decision Tree based ID3 algorithm. **Problem Description:** Upload the Iris Data set that is available in the scikit-learn library, preprocess the data by splitting the data into features and labels. Train the Decision Tree to build the decision tree model. Test the model using the test set and print the accuracy**.** **Method:** The method utilized here is supervised machine learning algorithm called Decision Tree Learning and apply this knowledge to classify a new sample. **Theory Reference:** Module 3	2
Program 5 **Title:** Implement Leave-One-Out Cross-Validation **(**LOOCV) using the Iris dataset for finding the accuracy for each fold. **Problem Description:** 1. Split the dataset into N subsets (where N is the number of samples). 2. For each subset, use the data points in the other N-1 subsets to train the model, and the data point in the current subset as the test set. 3. Evaluate the model's performance for each of the N iterations. 4. Calculate the overall performance (e.g., accuracy, error) by averaging the results of the N iterations. **Method:** LOOCV is a powerful cross-validation technique, especially for small datasets,	2

where every data point matters. Implement LOOCV using Scikit-learn in Python, and the technique can be applied to any machine learning model for model evaluation. **Theory Reference:** Module 3	

Program 6 **Title:** Implement Principal Component Analysis (PCA) for reducing the dimensionality of the Iris dataset from 4 features to 2. **Problem Description:** 1. Load the Iris dataset. 2. Standardize the data (mean = 0, variance = 1). 3. Apply PCA to reduce the data from 4 features to 2. 4. Visualize the reduced 2D data. **Method:** PCA helps in visualizing high-dimensional data in a lower-dimensional space while preserving most of the information. The plot should show the 2D projections of the Iris data on the first two principal components. You should see that the classes (species) are somewhat separated in the 2D space. **Theory Reference:** Module 3	2
Program 7 **Title:** Implement and demonstrate the working of Linear SVM algorithm. **Problem Description:** 1. **Find the Optimal Hyperplane**: SVM tries to find the hyperplane (decision boundary) that maximizes the margin between the closest data points of different classes **Support Vectors**: The data points closest to the hyperplane are called support **vectors**. These points are crucial in defining the optimal hyperplane. 3. **Kernel Trick**: If the data is not linearly separable, SVM can use the kernel trick to map the data into a higher-dimensional space where it becomes linearly separable. 4. **Classification**: Once the hyperplane is defined, it can be used to classify new data points. **Method: Support Vector Machine (SVM)** is a powerful algorithm used for classification and regression tasks, though it is more commonly used for classification. SVM works by finding the hyperplane that best separates the data into different classes while maximizing the margin between the classes. **Theory Reference:** Module 3	2

Program 8 **Title:** Evaluate the performance metrics of Linear SVM algorithm **Problem Description:** 1. Load the Iris dataset. 2. Split the dataset into training and testing sets. 3. Train a classification model (Support Vector Machine in this case). 4. Make predictions using the trained model. 5. Evaluate the model using the common evaluation metrics: Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. **Method:** By evaluating a model using multiple metrics for the Iris dataset, the SVM model should achieve perfect accuracy, precision, recall, and F1 score, which indicates that the model performs well. **Theory Reference:** Module 3	2
Program 9 **Title:** Demonstrate the working model of K-means clustering algorithm with Expectation Maximization Concept **Problem Description:** K-means clustering algorithm with the Expectation- Maximization (EM) concept on a popular dataset called as Iris dataset. The Iris dataset is a commonly used dataset for clustering and classification tasks. It contains 150 samples of iris flowers, each with four features (sepal length, sepal width, petal length, and petal width), and it is widely used in machine learning for clustering tasks. **Method:** The method utilized here is unsupervised machine learning algorithm for clustering using Iris Data set. Apply EM algorithm to cluster a set of data stored in a .CSV file. Use the same data set for clustering using k-Means algorithm. Compare the results of these two algorithms and comment on the quality of clustering. You can add Python ML library classes/API in the program. **Theory Reference:** Module 5	2
Program 10 **Title:** Implement and demonstrate DBSCAN clustering on a real social media dataset. **Problem Description:** Use the Twitter API to collect geospatial data (latitude and longitude) of tweets, and then apply DBSCAN clustering to group tweets by location. Visualize the clusters and detect any noise points (tweets that do not belong to any cluster). **Method:** DBSCAN is a great algorithm for clustering geospatial data from social media platforms like Twitter, especially when the clusters may have arbitrary shapes and when some posts may not belong to any cluster. **Theory Reference:** Module 5	2

Suggested Learning Resources:	
Textbooks:	
1	Andreas C. Müller, Sarah Guido, Introduction to Machine Learning with Python**,** O'Reilly Media, Inc, 2016.
Reference Books:	
1	Tom Michel, Machine Learning, McGrawHill Publication, First Edition, 2013.
2	Aurelien Geron, Hands-On Machine Learning with Scikit-Learn and TensorFlow- Concepts, Tools and Techniques to Build Intelligent Systems, Third Edition, O'Reilly, 2023.

Web links and Video Lectures (e-Resources):

- https://nptel.ac.in/courses/106105152
- https://nptel.ac.in/courses/106106139

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning

- Real world problem solving for Machine Learning using Checker board game.
- Classification of text for spam email detection, Topic categorization and sentimental analysis

Course outcomes:	
CO1	Apply machine learning algorithms to create ML models.
CO2	Analyze the performance of different machine learning models created using specific datasets
CO3	Develop machine learning models for real world problems
CO4	Use machine learning tools to develop machine learning models

CO-PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	2	-	-	-	-	-	-	-	-	3	-	-
CO4	-	-	-	-	3	-	-	-	-	-	-	-	-	-

Course Title	Software Engineering	**Semester**	VI
Course Code	CS353	**CIA**	50
Total No. of Contact Hours	(40 Th+20 Pr) 60hrs	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:2	**Total**	100
Credits	4	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview: This course provides a comprehensive introduction to software engineering principles, methodologies, and best practices for designing, developing, testing, and maintaining software systems. It covers the entire software development lifecycle (SDLC), including requirement analysis, system design, coding, testing, deployment, and maintenance. The course also explores agile methodologies, software quality assurance, version control, and emerging trends in software engineering.

Course objective is to:

1. Provide clear understanding of Software development process.
2. Introduce new trends in agile software development process.
3. Demonstrate the design patterns of different types of systems.
4. Understand the testing process for quality software.

Prerequisite Knowledge: Students must be familiar with application development framework.

Module-1: **Introduction**	**8 Hours**

Introduction: Professional software development, Software engineering ethics, Attributes of good software.

Software Process & Agile Software Development:

Software Process models: waterfall, incremental development, reuses oriented, Process activities.

Agile methods, Plan-driven and agile Development, Extreme Programming, Agile project management, scrum.

Activity: Form Pairs of students and follow XP for assigned simple coding tasks.

Text Book 1: Chapter 1, 2, 3 **Text Book 2**: 2.4.4

Module-2: Requirement Engineering	**8 Hours**

Functional and Non-functional requirements, The software requirements document, Requirements specification, Requirements engineering processes, Requirement elicitation and analysis, Requirements validation, Requirements management.

Case Study 1: Requirement gathering for Healthcare Management Systems.

Activity: The case studies must end with the presentation/submission of the requirement document.

Textbook 1: Chapters 4

Textbook 2: Case Study 4.1

Module-3: System Modelling and Architecture	**8 Hours**

System Modelling and Architectural Design: Context models, Interaction models, Structural models, Behavioural models, Model-driven engineering, **Software Architecture**: The role of software architecture, architectural views, Architectural patterns. **Text Book 1**: Chapter 5, 6	
Module-4: Software Design	**8 Hours**
Design and Implementation Software Design: Object-oriented design using the UML, Design patterns, Implementation issues, UML Diagrams, Use Case Model. **Distributed software engineering:** Distributed system issues, Client-server computing, Architectural patterns for distributed systems, Software as a service. **Activity:** Design the client-server system using design patterns **Text book 1:** Chapters 7**, Textbook 2:** 7.3, 7.4	
Module-5: Software testing and Quality management	**8 Hours**
Software testing: Introduction to Manual Testing, Testing process, Levels of Testing: Unit testing, integration testing, White Box Testing & Its Types, Black-box testing **Quality Management:** Software quality, Reviews and inspections: The review process, Software measurement and metrics: **Activity**: Explore selenium tools for automated testing. **Text Book 1**: Chapter 8, 24	
Follow-on Courses: Advanced Software Engineering, Software Testing & Quality Assurance, DevOps, Software project management.	

Practice Experiments	
1	Study the Agile process of software development • Scrum • XP • Lean software development • TDD • Kanban
2	**Title :** Design and implement web application in a short sprint using scrum framework. **Problem Description:** Students are expected to develop a 2 to 3 page web/ mobile application by forming the team of 5 members. The roles are to be decide by team members: • Product owner- course instructor/ 1 student from team. • Scrum master- manages the process, conduct meetings • Development team- Build and deliver product. The Scrum process has to be managed using the agile tool – jira /Trello. Free versions.
3	**Title:** To practice Pair Programming through writing simple code. **Problem Description:** Students are expected to for pair in which one writes the code and other improve the quality. Next hour students are expected to exchange the role. The following programs can be used to practice PP • Menu driven program to implement stack operation • To find GCD of the two numbers • To check whether a given string is palindrome. • Pattern matching program. • Find a pair of strings of numbers that do not exceed a given limit.

4	**Title** : Practice TDD(Test-driven development for the simple calculator application. **Problem description**: Students are expected to write small part of the calculator program ex-adding functionality. They should run the code to test the add functionality using testing frameworks like Pytest, Junit test code parallelly.Next functionality is parallelly developed and tested.
5	Title: Practice kanban agile methodology **Problem Description**: students are expected to design either a physical or software based Kanban board for the given sample problems. ● Team board for online marketing. ● Team board for Sales ● team board for product development ● Team board for time driven workflow. ● Team board event driven workflow. ● HR Kanban board Board for content creation
6	**Mini Project:** For any problem selected, students must follow agile software development process and demonstrate it with the report.

Textbooks:	
1	Ian Sommerville : Software Engineering, 10th edition, Person Education Ltd, 2016
2	Rajib Mall :Fundamentals of Software Engineering, Fifth Edition, PHI Learning, 2018.
Reference Books:	
1	Roger S Pressman, Bruce R Maxim, Software Engineering: A Practitioner's Approach, Seventh Edition, McGraw-Hill Education, 2009.

Web links and Video Lectures (e-Resources):

https://onlinecourses.nptel.ac.in/noc20_cs68/preview

https://onlinecourses.nptel.ac.in/noc22_cs61/preview

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning

1. Develop a Software Application(Web/ Mobile application) and apply concepts like requirements gathering, design, development, and testing to build a complete application.
2. Learn the usage of tools like Github, Jira, for project management, version control.
3. Understand usage of JUnit (for Java), pytest (for Python), or Mocha (for JavaScript) for testing.
4. Can choose 6 case study (user stories), form 6 groups of 10 students and plan 5 assignments as follows;
5. Requirement gathering
6. Choosing proper software process model
7. Draw architecture diagrams.
8. Use one of design and comment on suitable design Patterns
9. Perform testing using software tool
10. At the end of course each team will submit a report.

Course outcomes: At the end of the course, students will be able to:	
CO1	Apply suitable software design principles, models, and architectures to develop software solutions.
CO2	Analyse software requirements and prepare appropriate requirement specification document(SRS).
CO3	Use software engineering tools and techniques for modeling, implementation, and documentation of software systems.
CO4	Demonstrate teamwork, ethical practices, and professional responsibilities in software engineering projects.

CO-PO Mapping

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3													
CO2		3									3		2	
CO3					3									
CO4								2	2					

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Compiler Design	**Semester**	VI
Course Code	CS354	**CIA**	50
Total No. of Contact Hours	(40 Th+20 Pr) 60hrs	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:2	**Total**	100
Credits	4	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview: This course covers the fundamental principles, methodologies, and tools used to build and construct compilers. Students will learn how to translate high-level programming languages into machine code, as well as how contemporary compilers work.

Course objective is to:

1. Present the principal domains of language translation and compiler design.
2. Enhance understanding of the many stages of a compiler and its applications.
3. Demonstrate intermediate code using technique of syntax directed translation.
4. Illustrate the various optimization techniques for designing various optimizing compilers.

Prerequisite Knowledge: Students should have a solid understanding of Formal Language & Automata Theory, Discrete Mathematics, Knowledge of C/C++, Java for compiler implementation.

Module-1 **8 Hours**

Introduction to Compiling and Lexical Analysis

Introduction, Language Processor, The Structure of Compilers, Evolution of programming Languages, Application of Compiler Technology. Lexical Analysis-The Role of Lexical Analyzer, Input Buffering, Specification of Tokens, Recognition of Tokens.

Case Study: Investigate how finite automata are utilized in the lexical analysis phase of compiler design for token recognition, error detection, and pattern matching. The case study focuses on creating and studying a finite automaton that identifies tokens such as keywords, identifiers, numeric literals, and operators in a hypothetical programming language.

Textbook 1: Ch 1.1-1.3, 1.5, 3.1-3.4

Module-2 **8 Hours**

Syntax Analysis

Introduction, Context Free Grammars, Writing a Grammar, Top-Down Parsing, Bottom-Up Parsing, Introduction to LR Parsing: Simple LR, More Powerful LR Parsers.

Case Study: Investigate how syntax analysis was implemented and optimized in a custom C compiler, tackling issues such as unclear grammar, error handling, and performance bottlenecks.

Textbook 1: Ch 4.1 to 4.7

Module-3 **8 Hours**

Lex and Yacc- The Simplest Lex Programs, Grammars, Parser-Lexer Communication, A YACC Parser, The Rules Section, Running LEX and YACC, LEX and Hand- Written Lexers, Using LEX - Regular Expression.

Using YACC – Grammars, Recursive Rules, Shift/Reduce Parsing, What YACC Cannot Parse, A YACC Parser - The Definition Section, The Rules Section, The LEXER.

Syntax-Directed Translation
Syntax-Directed Definitions, Evaluation orders for SDD, Application of Syntax Directed Translation**. Case Study:** Demonstrate how Syntax-Directed Translation (SDT) provides efficient expression evaluation and intermediate code creation while parsing. By including semantic actions into YACC, the compiler effectively transformed arithmetic expressions into optimized TAC while ensuring accurate precedence, efficient memory utilization, and strong error handling.

Textbook 2: Ch 1, 2 & 3
Textbook 1: Ch 5.1 to 5.3

Module-4	**8 Hours**

Intermediate Code Generation
Variants of Syntax trees, three address code, Types and Declaration-Type Expressions, equivalence, Declaration, Control flow.

Case Study: Intermediate Code creation (ICG) is a critical phase of a compiler that converts high-level source code to an intermediate representation (IR), which acts as a link between front-end parsing and back-end code creation. The efficiency of this intermediate code has a significant impact on the performance, memory utilization, and execution speed of the final compiled program. Implement important optimization approaches to increase efficiency.

Textbook 1: Ch 6.1 to 6.6

Module-5	**8 Hours**

Code Generation and optimization
Issues in the design of Code Generator, The Target Language, Address in the target Code, Basic Blocks and Flow graphs, Optimization of Basic blocks, A Simple Code Generator, Peephole Optimization.

Case Study: Develop a compiler for a simple Mini-Language (MiniLang), which includes basic programming constructs such as variable declarations, arithmetic operations, conditional statements, and loops. The compiler should generate an Intermediate Representation (IR) in the form of Three- Address Code (TAC) and produce assembly-like target code for a virtual machine.
Textbook 1: Ch 8.1 to 8.7

Practice Experiments:
Program 1:
Title: Write a C program to identify different types of tokens in a given Program.
Problem Description: Identify and classify different types of tokens in a given C program. Tokens are the basic building blocks of any program and include identifiers, keywords, constants, operators, special symbols, and punctuation.
Method: To identify tokens in a program, define patterns using regular expressions to represent categories such as keywords, identifiers, and operators. Tokenize the input by splitting it into individual elements, then match each token against the defined patterns to classify its type. Finally, return the categorized tokens along with their corresponding types.
Theory Reference: Module 1

Program 2:
Title: Write a Lex Program to implement a Lexical Analyzer using Lex Tool.
Problem Description: The program should read a source code file written in C and perform lexical analysis by identifying and classifying the tokens present in the code. Tokens are the fundamental elements of a program, such as keywords, identifiers, operators, constants, and special symbols.
Method: Lex is a tool used to create lexical analyzers by defining token patterns and their actions in a Lex file. It generates a C program that processes input to identify and categorize tokens based on the specified rules. The program is compiled and executed to perform lexical analysis efficiently.
Theory Reference: Module 1 & 3

Program 3:
Title: Write a C program to simulate Lexical Analyzer to validating a given input string.
Problem Description: Simulate a Lexical Analyzer to validate a given input string. The program should identify and classify the tokens present in the string based on their types, ensuring the input string follows valid lexical rules. The lexical analyzer must process the string character by character, identify tokens, and validate the input.
Method: A lexical analyzer validates an input string by defining token patterns using regular expressions, matching the string against these patterns, and categorizing each token as valid or invalid based on the matches.
Theory Reference: Module 2

Program 4:
Title: Write a C program to implement a Recursive Descent parser.
Problem Description: The Recursive Descent Parser should validate whether an input string belongs to the language generated by the grammar. This parser is a top-down parser and uses recursive procedures for each non-terminal in the grammar.
Method: Create a function for each non-terminal in the grammar that parses its corresponding rule and design the functions to consume tokens and call each other recursively to match the input against the grammar.
Theory Reference: Module 2

Program 5:
Title: Write a C program to compute First and Follow sets for given Grammar.
Problem Description: compute the FIRST and FOLLOW sets for a given context-free grammar (CFG). The program should take the grammar as input (in a standard form) and calculate the FIRST and FOLLOW sets for each non-terminal in the grammar. These sets are critical for constructing predictive parsers such as LL(1) parsers.
Method: To compute the First set, identify the terminals or ε derivable from each non-terminal by processing its productions. To compute the Follow set, determine the terminals and non-terminals that can appear immediately after each non-terminal, using propagation rules for ε. Repeat the process for both sets until no changes occur.
Theory Reference: Module 2

Program 6:
Title: Write a C program for eliminating the left recursion and left factoring of a given grammar.
Problem Description: Eliminate left recursion and perform left factoring for a given context-free grammar (CFG). The program should take the grammar as input, identify any left recursion and/or common prefixes in production rules, and modify the grammar to remove left recursion and perform left factoring where necessary.
Method: Replace productions of the form $A \rightarrow A\alpha \mid \beta$ with $A \rightarrow \beta A'$ and $A' \rightarrow \alpha A' \mid \varepsilon$, and for indirect left recursion, substitute dependent rules first; for left factoring, factor out common prefixes in productions like $A \rightarrow \alpha\beta1 \mid \alpha\beta2$ by rewriting them as $A \rightarrow \alpha A'$ and $A' \rightarrow \beta1 \mid \beta2$.
Theory Reference: Module 4 & 5

Program 7:

Title: Write a C program to check the validity of input string using Predictive Parser.

Problem Description: The program should implement a Predictive Parsing Table for a given grammar and use it to parse the input string step-by-step to determine whether it is valid (belongs to the language generated by the grammar).

Method: Initialize a stack with the start symbol and $, then parse the input string by matching or expanding symbols using the table. If the stack is empty and the input is fully consumed, the string is valid; otherwise, it is invalid.

Theory Reference: Module 2

Program 8:

Title: Write a C program for generating the three-address code of a given expression/statement.

Problem Description: The program should take an expression as input and output its corresponding three-address code representation. The three-address code should include intermediate variables and operations to represent the input in a form suitable for further optimization and code generation in a compiler.

Method: Breaking expression or statement it into simpler instructions with at most one operator and three operands. The program must respect operator precedence, associativity, and parentheses while producing intermediate code. For example, the input a = b + c * d would output t1 = c * d, t2 = b + t1, and a = t2.

Theory Reference: Module 4 & 5

Follow on Courses: Code Generation for Modern Architectures, AI and Machine Learning in Compiler Design

Suggested Learning Resources:

Textbooks:

1	Alfred V Aho, Monica S.Lam, Ravi Sethi, Jeffrey D Ullman, Compilers- Principles, Techniques and Tools, 2nd Edition, Pearson Education,2013.
2	Doug Brown, John Levine, Tony Mason, lex & yacc, 2nd Edition, O'Reilly Media, October 2012.

Reference Books:

1	Kenneth C Louden; Cengage Learning, Compiler Construction Principles & Practice, 1st Edition, 2009.
2	Charles N. Fischer, Richard J. leBlanc, Jr., Crafting a Compiler with C, 1st Edition, Pearson Education, 2009.

Web links and Video Lectures (e-Resources):

1. https://archive.nptel.ac.in/courses/106/105

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning

- Develop a clear understanding of **lexical rules** and how tokenization works in compilers.
- Generate **Three-Address Code (TAC)** for expressions and statements.

Course outcomes: At the end of this course, Students will be able to	
CO1	Apply the fundamental concepts of compiler design to analyze the structure and phases of a compiler.
CO2	Analyze syntax analysis techniques using context-free grammars and parsing methods in compiler design.
CO3	Develop a solution for parsers using LEX and YACC for intermediate code generation.
CO4	Use engineering tools lex & Yaac to solve compiler-related problems.
CO5	Demonstrate the proficiency in designing, implementing, and testing the compiler components.

CO-PO Mapping

CO / PO	P O 1	PO 2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO1	3													
CO2		3									2			
CO3			1											
CO4					3									
CO5								2	2					

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Research Methodology and IPR	**Semester**	VI
Course Code	RM355	**CIA**	50
Total No. of Contact Hours	25	**SEE**	50
No. of Contact Hours/week (L:T:P)	2:0:0	**Total**	100
Credits	2	**Exam. Duration**	3 Hours
Teaching Department(s)	All Engineering Departments		

Course objective is to:

1. Give an overview of the research methodology and explain the technique of defining a research problem.
2. Carry out a literature search, its review, developing theoretical and conceptual frameworks and writing a review.
3. Explain various research designs and their characteristics.
4. Explain the concept of Intellectual Property with its various types.
5. Understand the ethics in Research and IP.

Module-1	**5 Hours**

Research Methodology: Introduction, Meaning of Research, Objectives of Research, Types of Research, Research Approaches, Significance of Research, Research Methods versus Methodology, Research and Scientific Method, Research Process, Criteria of good research, Common Issues Encountered by Researchers in India. Typical research outcomes – like research publications, IP, research proposals.

Module-2	**5 Hours**

Defining the Research Problem: Reviewing the literature: significance of the literature review, literature-review process, searching the existing literature, reviewing the selected literature, writing about the literature reviewed. Developing a conceptual framework, bringing clarity and focus to your research problem, Research Problem, Selecting the Problem, Necessity of Defining the Problem, Technique Involved in Defining a Problem.

Activity: Literature review – finding research gaps – defining a problem statement using widely-used bibliographic database(s).

Module-3	**5 Hours**

Research Design: Meaning of Research Design, Need for Research Design, Features of a Good Design, Important Concepts Relating to Research Design, Basic Principles of Experimental Designs, Important Experimental Designs.

Attributes and Reasons of Technical Writing, Patent or research Paper? - The Choice, Writing Strategies. Research Publication: Structure and Approach, Title, Abstract, Introduction, Methods, Results and Discussions, Table, Figures, Acknowledgments, and Closures, Language Skills, Writing Style, and Editing

Activity: writing of a research paper

Module-4	**5 Hours**

Introduction To Intellectual Property:

Intellectual Property (IP) Acts: different types of IPs and its importance, Patent Acts: Indian patent acts 1970. Design Act: Industrial Design act 2000. Copyright acts: Copyright Act 1957. Trade Mark Act 1999.

Types of patents, To Patent or Not to Patent an Invention, Conditions for Obtaining a Patent Protection, Rights Associated with Patents, Inventions Eligible for Patenting, Non-Patentable Matters.

Need for a Patent Attorney / Agent, Avoid Public Disclosure of an Invention before Patenting. Process of Patenting: Pre-grant Opposition, Examination, Grant of a Patent.

Activity: Patent search, exploring if a given idea is patentable or not.

Module-5	5 Hours
Jurisdiction of Filing Patent Application, Commercialization of a Patent, Patent Infringements. Process of disclosing an invention and typical components of the same. Case Studies on Patents. Case study of Curcuma (Turmeric) Patent, Case study of Neem Patent, Case study of Basmati patent. IP Organizations in India. Copyrights of the Author: Copyright Registration, Copyright Infringements, Validity of Copyright, Copyright and the word 'Publish', Transfer of Copyrights to a Publisher, International Copyright Agreements. Ethics in Research and IP: Ethics in Engineering Research Practice, Types of Research Misconduct, Ethical Issues Related to Authorship. Activity: Plagiarism checking for given written material.	

Suggested Learning Resources: **Textbooks:**	
1	C.R. Kothari, Gaurav Garg, Research Methodology: Methods and Techniques, 4th Edition, New Age International ,2018.
2	Ranjit Kumar, Research Methodology a step-by- step guide for beginners. (For the topic Reviewing the literature under module 2), SAGE Publications Ltd 3rd Edition, Study Material, 2011.
3	Debirag E. Bouchoux, Intellectual property, Cengage learning, 2013.
Reference Books:	
1	Trochim, Research Methods: the concise knowledge base, Atomic Dog Publishing, 2005.
2	Fink A, Conducting Research Literature Reviews: From the Internet to Paper, Sage Publications, 2009.
3	Dipankar Deb,Rajeeb Dey and Valentina E. Balas, Engineering Research Methodology A Practical Insight for Researchers, Springer Publications,2019.
Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning The students with the help of the course teacher can take up technical activities which will enhance their skill or the students should interact with industry (small, medium and large), understand their problems or foresee what can be undertaken for study in the form of research/testing/projects, and for creative and innovative methods to solve the identified problem. The prepared report shall be evaluated for CIE marks.	
Course outcomes:	
CO1	Identify the suitable research methods and articulate the research steps in a proper sequence for the given problem.
CO2	Plan literature survey, define the problem statement and suggest suitable solution for the given problem and present in the format of the research paper (IEEE).
CO3	Analyse the problem and conduct experimental design with the samplings.
CO4	Plan patent search, analyse if an invention is patentable or not.
CO5	Apply some concepts/section of Copy Right Act /Patent Act /Cyber Law/ Trademark to the given case and develop–conclusions.

CO PO Mapping:														
CO/PO	PO1	PO2	PO3	PO 4	PO 5	PO 6	PO 7	PO8	PO 9	PO1 0	PO1 1	PS O1	PS O2	PS O3
CO1	1	2	-	-	-	-	-	2	-	-	-	-	-	-
CO2	1		3	-	-	-	-	2	-	-	-	-	-	-
CO3	-	2	-	3	-	-	-	3	-	-	-	-	-	-
CO4	1	-	3	-	-	-	-	3	-	-	-	-	-	-
CO5	1	-	3	-	-	-	-	2	-	-	-	-	-	-

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Innovation and Entrepreneurship	**Semester**	VI
Course Code	CS361	**CIA**	50
Total No. of Contact Hours	25	**SEE**	50
No. of Contact Hours/week	2:0:0	**Total**	100
Credits	2	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview: The Innovation and Entrepreneurship course aims to provide students with the information, mindset, and abilities needed to discover opportunities, develop new solutions, and establish successful companies. It teaches the principles of creativity, business model development, and entrepreneurial thinking, promoting a proactive approach to problem resolution and value creation in a dynamic market setting.

Course Objective is to:

- Gain a clear understanding of innovation, creativity, and entrepreneurship and their role in driving economic and societal growth.
- Cultivate critical thinking, problem-solving, and decision-making skills essential for entrepreneurial success.
- Learn about different types of innovation (product, process, business model, and social) and how to apply them to real-world challenges.
- Equip students with tools and techniques to identify, analyze, and evaluate business opportunities in various industries.

Pre-requisites: Basic understanding of how businesses work, Knowledge of budgeting, cash flow, and financial projections.

Module-1	**Hours: 5**

Introduction: Systematic Entrepreneurship, Purposeful Innovation and the Seven Sources for Innovative Opportunity, Detailed study of each source of innovation
Text Book1: Ch 1-5

Module-2	**Hours: 5**

The Practice of Innovation: Industry and Market Structures, Demographics, Changes in Perception, New Knowledge

Text Book1: Ch 6-9

Module-3	**Hours:5**

Entrepreneurship in the Service Sector: The Bright Idea, Principles of Innovation, Entrepreneurial Management, The Entrepreneurial Business

Text Book1: Ch 10-13

Module-4	**Hours: 5**

Entrepreneurial Strategies: Entrepreneurship in the Service Institution, The New Venture, Fustest with the Mostest, Hit Them Where They Ain't
Text Book1: Ch 14-17

Module-5	**Hours: 5**
Shifting Characteristics and Values: Ecological Niches, Changing Values and Characteristics, Conclusion: The Entrepreneurial ***Society*** Text Book1: Ch 18-19	
Follow on Courses: Focuses on early-stage startups and Help entrepreneurs create competitive innovation strategies.	
Suggested Learning Resources: **Textbooks:**	
1.	Peter F Drucker, Innovation and Entrepreneurship Practice and Principals, Harper &Row, Publishers, Inc. published 1993.
Reference Books:	
1.	John Heywood, Engineering Education: Research and Development in Curriculum and Instruction, Wiley publications, 2005 .
2.	Heinz Weihrich, Mark V Cannice, Harold Koontz, Management: A Global and Entrepreneurial Perspective, 13th edition, Tata McGraw Hill, 2011.
Web links and Video Lectures (e-Resources): • https://archive.nptel.ac.in/courses/110/107	
Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning The students with the help of the course teacher can take up relevant technical –activities like presentation, Role-Playing Scenarios and Collaborative Learning Projects which will enhance their skills. The prepared report shall be evaluated for CIE marks.	

Course outcomes:	
CO1	Analyze industry and market structures to identify and evaluate opportunities for innovation.
CO2	Develop innovative entrepreneurial ideas and business strategies applicable to service sectors and new venture creation
CO3	Collaborate effectively in teams to conceptualize, evaluate, and refine entrepreneurial strategies and venture ideas through group discussions and activities.
CO4	Communicate entrepreneurial ideas, innovation strategies, and business concepts clearly and professionally through presentations, reports, and discussions.

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1		2												
CO2			2											
CO3								2						
CO4									2					

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Active Learning	**Semester**	VI
Course Code	CS362	**CIA**	50
Total No. of Contact Hours	25	**SEE**	50
No. of Contact Hours/week (L:T:P)	2:0:0	**Total**	100
Credits	2	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course overview:

This course introduces educators to active learning strategies and their application in computer science classrooms. Through interactive teaching techniques such as think-pair-share, simulations, and collaborative problem-solving, instructors will learn how to engage students, enhance critical thinking, and improve learning outcomes.

Course objective:

1. Develop proficiency in applying active learning strategies to foster student engagement and problem-solving skills in computer science.
2. Enhance students' ability to analyze and solve complex computer science problems through collaborative work, critical thinking, and the use of technological tools.
3. Apply varied teaching methods and assessments to accommodate different learning levels and goals, promoting inclusive and effective learning in computer science.

PREREQUISITE KNOWLEDGE: This course requires that the students are familiar with fundamentals of C, Design and analysis of Algorithms.

Module-1	**5 Hours**

What is active learning?, active learning strategies: Purposeful Pause, Quick Write or "Minute" Paper, Think-Pair-Share (TPS) ,Polling/Peer Instruction ,Concept Map, Case Study/Group Problem Solving ,Think-Aloud Problem Solving, Jigsaw. Gallery Walk

Activity: Present students with a real-world problem related to computer science and ask them to work in small groups to analyze and solve it. This can involve debugging code or designing a new system.

Resources:

a) https://gsi.berkeley.edu/gsi-guide-contents/learning-theory-research/cognitive-constructivism/
b) https://gsi.berkeley.edu/gsi-guide-contents/learning-theory-research/social-constructivism
c) https://teaching.berkeley.edu/active-learning-designs

Module-2	**5 Hours**

What can active learning look like in practice?,How to explain to your students why you are using clickers, Evidence-based practices for using clickers and clicker discussions.

Activity: Present a real-world problem, such as designing a system to manage customer orders in a store. Ask students to work in groups to come up with a solution using concepts like databases, object-oriented design, and algorithms.

Resources: https://seercenter.uga.edu/realisevideos_howto/

https://www.ibiology.org/playlists/scientific-teaching-series/
https://onehe.org/equity-unbound/

Module-3	**5 Hours**
How use to think-pair-share in your classroom, Step-by-step guide to creating and using clicker questions, How to use simulations to probe and challenge students' thinking in class. **Activity:** Start by crafting a question that prompts critical thinking about the material you've just taught. Ensure the question is open-ended and requires students to explain their reasoning. **Resources**: https://teaching.berkeley.edu/teaching-guides-resources/teaching-your-course/active-learning	
Module-4	**5 Hours**
Integrating simulations and clicker questions to get students thinking**, Teaching a Mixed Level Disparate Class**- Do Students Need To Have a Pre-Determined Baseline of Knowledge For Your Course?. **Activity:** Use an algorithm visualizer such as Algorithm Visualizer or Sorting Algorithm Visualizer. This tool will allow students to visualize the behavior of different sorting algorithms (e.g., Bubble Sort, Merge Sort, Quick Sort) with various data inputs **Resources**: • https://teaching.berkeley.edu/teaching-guides/running-your-course/teaching-mixed-level-disparate-class	
Module-5	**5 Hours**
How Much Variety Can You Provide In Your Teaching and Assessments?, Are You Aware Of Your Students' Potentially Disparate Goals?, Can You Leverage Group Work? **Activity:** **Tool**: Choose a tool like Replit or Jupyter Notebooks, where students can collaboratively write and test code in real-time, allowing them to experiment with algorithms in different programming languages. **Varied Problem Sets**: Prepare different levels of algorithmic challenges based on complexity: • **Beginner Level**: Sorting algorithms (e.g., Bubble Sort, Insertion Sort) • **Intermediate Level**: Divide and conquer algorithms (e.g., MergeSort, QuickSort) • **Advanced Level**: Graph algorithms (e.g., Dijkstra's algorithm, A* search) Have students work together to write the code for the assigned algorithm. As they work, encourage them to discuss their thought processes, troubleshoot together, and explain concepts to one another. **Resources**: • https://teaching.berkeley.edu/teaching-guides/running-your-course/teaching-mixed-level-disparate-class	

Suggested Learning Resources:

Web links and Video Lectures (e-Resources):

Suggested Learning Resources (Web Links & Video Lectures):

a. **Cognitive Constructivism – Berkeley GSI Guide**
b. **Social Constructivism – Berkeley GSI Guide**
c. **Active Learning Designs – Berkeley Teaching Guide**
d. **Active Learning Overview (Section 0)**
e. **Using Active Learning Strategies (Section 1)**
f. **Think-Pair-Share and Other Tools (Section 2)**
g. **Polling and Peer Instruction (Section 3)**
h. **Concept Mapping and Group Work (Section 4)**
i. **Case Studies and Problem Solving (Section 5)**
i. **Simulations in Teaching (Section 6)**

1. **Teaching a Diverse Classroom (Section 7)**

   Assessment and Feedback (Section 8)

   REALISE Videos – SEER Center

   Scientific Teaching Series – iBiology

1. **Equity Unbound – OneHE**

   Teaching Mixed-Level Classes – Predetermined Baseline

   Teaching Mixed-Level Classes – Variety

   Teaching Mixed-Level Classes – Disparate Goals

   Teaching Mixed-Level Classes – Group Work

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning

1. Real world problem solving using group discussion.

Course outcomes:	
CO1	Summarize the theoretical foundations of active learning
CO2	Execute technology-aided instruction and interactive simulations to evaluate student comprehension in real-time.
CO3	Solve real-world algorithmic problems by facilitating collaborative coding environments and peer-instruction activities
CO4	Formulate varied instructional designs and assessment rubrics tailored for mixed-level classrooms to support disparate student goals.

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2													
CO2		2												
CO3								2						
CO4									2					

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

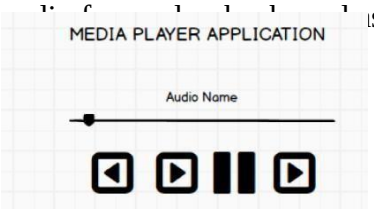
Course Title	Mobile Application Development Laboratory	**Semester**	VI
Course Code	CS363	**CIA**	50
L:T:P	1:0:2	**SEE**	50
Hrs. / Week	2	**Total**	100
Credits	2	**Exam Duration**	3 Hours
Teaching Department	CS		

Course objectives :	
1	Learn and acquire the knowledge of Android Programming.
2	To develop android application with responsive User Interface
3	Create, modify and query on SQlite database.
4	Test and debug a mobile application for a reliable output.
5	To explore various concepts of android application.

Pgm. No.	**List of Experiments / Programs**	**Hours**	**COs**
Prerequisite Knowledge: Students must be proficient in Java/ Kotlin Programming language			
	PART-A		
1	**Program 1:** **Title:** Design and develop a calculator application to perform basic arithmetic operations. **Problem Description:** The problem involves creating the calculator application with good user interface to perform basic calculations. The program can be further extended for other math operations. **Method:** First step is to design UI using **TableLayout**. Second step is to implement a common method in the class to handle the click events. The labels of the button can be used to know the button clicked. The content of edit text is appended upon clicking the buttons. String operations are used to parse the edit text content to perform operation and display result. **Reference:** Google Developer Training, **"Android Developer Fundamentals Course –Concept Reference",** Google Developer Training Team, 2017. https://www.gitbook.com/book/google-developer-training/android-developer-fundamentals course-concepts/details	2	1
2	**Program 2:** **Title:** Create a secure SIGN- UP and SIGN-IN application with Username and Password. Validation of password should happen based on the following rules: Password should contain uppercase and lowercase letters. • Password should contain letters and numbers. • Password should contain special characters. • Minimum length of the password (the default value is 8) **Problem Description:** The problem involves creating two activities. In first activity password validation is done for the given rules. If validation is successful, second activity should be displayed. User try to login using username and password, this activity should verify the password and display appropriate message. Three attempts are given to user after which the button should be disabled. **Method:** In this program, the concept involved is **Intent.** The intent object is used to pass the data to the next activity. The password validation can be done using regular expressions. **Reference:** Google Developer Training, **"Android Developer**	2	1

	Fundamentals Course –Concept Reference", Google Developer Training Team, 2017.		

3	Program 3: Title: Devise a mobile application to set alarm clock. Problem Description: The problem involves alerting the user in the form of notification /sound/ message once the alarm goes off. The user sets the alarm using proper UI and close the app. Method: The program involves use of Broadcast receiver to receive the broadcasts from android system for the alarm. Alarm Manager API to set the alarm. Pending Intent for communication between the receiver and the activity for alert. UI is designed using TimePicker dialog. **Reference: https://developer.android.com/develop/background-work/services/alarms/schedule**	2	2
4	Program 4: Title: Devise application to parse XML/JSON file and display the data using appropriate UI. Problem Description: User should create the XML file with the data given, button interface and layout to display the data. Upon clicking the button, data has to be displayed in the layout. Method: In this program, XML file to be stored in res folder. Program can use DOM parser or XMLPullParser(recommended) to extract the data and update activity. **Reference: https://developer.android.com/training/data-storage/sqlite#java**	2	1
5	Program 5: Title: Build a mobile application that create, save, update and delete data in a database Problem Description: The application involves UI interface to input ex student data, three buttons to insert data,delete and display data in SQLite database. Method: In this program, SQlite package consists of the API required to perform operations on atabase are used. The UI consists of Edittexts and buttons. The data can be displayed in the Dialog box or the List view. **Reference: https://developer.android.com/training/data-storage/sqlite#java**	2	3
6	Program 6: Title: Create an application that reads and writes data from and to the SD card. Problem Description: Write a program to create an activity having a Text box, and also Save, Open and Create buttons. The user has to write some text in the Text box. On pressing the Create button the text should be saved as a text file in SDcard. On subsequent changes to the text, the Save button should be pressed to store the latest content to the same file. On pressing the Open button, it should display. the contents from the previously stored files in the Text box. If the user tries to save the contents in the Textbox to a file without creating it, then a toast message has to be displayed saying "First Create a File". Method: The program has to be implemented using File I/O packeges(Java/Kotlin). If using java, FileInputStream and FileOutputStream can be used to perform I/O operations. Referenece:https://developer.android.com/training/data-storage	2	3

7	**Program 7:** **Title:** Create an application to receive an incoming SMS which is notified to the user. On clicking this SMS notification, the message content and the number should be displayed on the screen. **Problem Description** Write a program to create activity with text area and appropriate UI which displays the SMS content. Firstly user should receive notification about SMS. User must select it. The experiment can be tested on the real device. **Method:** Use pending intent concept and Notification manager API to receive SMS. Disable the real SMS app to receive SMS so that our app can receive. **Reference**:https://developers.google.com/identity/sms retriever/overview	2	2,4
8	**Program 8:** **Title:** Design and develop media player application. **Program Description**: Create an application to demonstrate a basic media player that allows the user to Forward, Backward, Play and Pause an audio. Also, make use of the indicator in the seek bar to move the [illegible] s required. The UI can as follows:  **Method:** Consider a single media file and play it using application. Use asynchronous concept in android to develop app. **Reference:** https://developer.android.com/develop/background-work/background-tasks	2	5

Suggested Learning Resources: **Textbooks/ Reference Books:**	
1	Google Developer Training, Android Developer Fundamentals Course –Concept Reference, Google Developer Training Team, 2017
2	Wei-Meng Lee, Beginning Android Application Development, 1st edition, Wiley ,2012
Web links and Video Lectures (e-Resources): **Developers.android.com/reference/android** **https://developer.android.com/courses**	
Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning Mini Project	
Course Outcomes: At the end of this course, Students will be able to	
CO1	Apply fundamental concepts of mobile application development to design and implement basic mobile applications.
CO2	Design mobile user interfaces using layouts, widgets and event handling mechanisms.
CO3	Integrate data storage mechanisms such as files, databases, or cloud services within mobile applications.
CO4	Design and demonstrate a mini-project or case study addressing real-world problems using mobile application technologies.

CO-PO Mapping															
CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3														
CO2		3	1												
CO3															
CO4							2	2	2		2				

High-3,Medium-2,Low-1

Course Title	Project Phase 1	**Semester**	VI
Course Code	CS356	**CIA**	50
Total No. of Contact Hours	---	**SEE**	0
No. of Contact Hours/week (L:T:P)	0:0:4	**Total**	50
Credits	2	**Exam. Duration**	---
Teaching Dept	CS		

The Final Year Project – Phase 1 involves the initial planning and groundwork for the main project, carried out in consultation with the assigned project guide. During this phase, students are expected to select a relevant and innovative topic, conduct a detailed literature review, define the problem statement, and outline the project objectives, methodology, and tools to be used. A comprehensive project proposal must be prepared and presented through a seminar before a review committee for evaluation and feedback. This phase ensures that the student has a clear direction and is well-prepared to move into the development and implementation stage in Phase 2.

Course Title	Cryptography	**Semester**	VI
Course Code	CS371	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L: T: P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview:

The Cryptography course introduces students to fundamental security concepts, encryption techniques, and cryptographic protocols. It covers symmetric and asymmetric encryption, including algorithms like DES, RSA, and Diffie-Hellman. Students will learn about message authentication, digital signatures, key management, and practical security applications. Hands-on activities include implementing cryptographic functions, analyzing security protocols, and verifying digital signatures. The course equips students with the knowledge to secure data and communications in real-world scenarios.

The course objective is to:

1. To understand the basic concepts of computer security and cryptography.
2. To study and analyze symmetric and asymmetric encryption techniques.
3. To learn about message authentication, digital signatures, and key management.
4. To explore practical applications of cryptography and security protocols.

PREREQUISITE KNOWLEDGE: This course requires that the students are familiar with concepts of Computer Networks, Modular Arithmetic.

Module-1 Introduction To Security & Cryptography	**8 Hours**

Computer Security Concepts – The OSI Security Architecture – Security Attacks – Security Services and Mechanisms – A Model for Network Security – Symmetric Cipher Model - Substitution Techniques - Transposition Techniques.

Activity: Cipher – Decipher Challenge

Encrypt a given plaintext using a substitution or transposition cipher (e.g., Caesar or Rail Fence). Exchange ciphertext with another group and decrypt it. The first group to decrypt correctly wins.

Textbook 1: 1.1 – 1.5, 1.8, 3.1 – 3.3

Module-2 Block Ciphers And The Data Encryption Standard	**8 Hours**

Block Ciphers and the Data Encryption Standard - Traditional Block Cipher Structure - The Data Encryption Standard - A DES Example - The Strength of DES - Block Cipher Design Principles - Multiple Encryption and Triple DES - Electronic Codebook - Cipher Block Chaining Mode - Cipher Feedback Mode - Output Feedback Mode.

Activity: Block Cipher Simulation

Write a program that divides a given plaintext into fixed-size blocks (e.g., 8 bytes). Apply a simple mathematical operation (e.g., XOR with a key) to each block for encryption. Students can observe the effect of block size on the output ciphertext.

Textbook 1: 4.1 – 4.5, 7.1 – 7.5

Module-3 Public Key Cryptography & Hash Functions	**8 Hours**
Principles of Public-Key Cryptosystems - The RSA Algorithm - Diffie-Hellman Key Exchange - Elgamal Cryptographic System - Elliptic Curve Cryptography - Pseudorandom Number Generation Based on an Asymmetric Cipher - Applications of Cryptographic Hash Functions - Two Simple Hash Functions - Hash Functions Based on Cipher Block Chaining - Secure Hash Algorithm (SHA). **Activity: Hash Function Verification** Write a program to compute the hash of an input message using SHA-256 (using Python's hashlib). Students can change the message slightly and observe how the hash output changes completely (avalanche effect). **Textbook 1: 9.1 – 9.2, 10.1 – 10.2, 10.4 – 10.5, 11.1, 11.2, 11.4, 11.5.**	
Module-4 Message Authentication Code & Digital Signatures	**8 Hours**
Message Authentication Requirements - Message Authentication Functions - Requirements for Message Authentication Codes - Security of MACs - MACs Based on Hash Functions: HMAC - MACs Based on Block Ciphers: DAA and CMAC - Digital Signatures - Elgamal Digital Signature Scheme - Schnorr Digital Signature Scheme - NIST Digital Signature Algorithm - Elliptic Curve Digital SignatureAlgorithm **Activity: Create your own Digital Signature** Manually create a simple digital signature by hashing the plaintext, encrypting the hash with a private key, and sharing the signature along with the plaintext. Verify the signature using a public key and discuss the roles of hashing, encryption, and key pairs in ensuring authenticity and integrity. **Textbook 2:** 12.1 – 12.6, 13.1 – 13.5	
Module-5 Key Management And Ip Security	**8 Hours**
Symmetric Key Distribution Using Symmetric Encryption - Symmetric Key Distribution Using Asymmetric Encryption - Distribution of Public Keys - X.509 Certificates - Public-Key Infrastructure - Remote User-Authentication Using Symmetric Encryption – Kerberos - Remote User Authentication Using Asymmetric Encryption - IP Security Overview - IP Security Policy - Encapsulating Security Payload **Activity: Verify X.509 Certificates** Students will be provided with a sample X.509 certificate and a public key. They will manually verify the certificate by comparing its hash and decrypting the signature using the public key. The activity will include a discussion on the role of X.509 certificates in establishing trust within a public-key infrastructure. **Textbook 2: 14.1 – 14.5, 15.1 – 15.4, 20.1 – 20.3**	

Advanced Learners:	
1	Study the impact of quantum computing on current cryptographic systems and explore post-quantum cryptography. https://www.coursera.org/courses?query=quantum+cryptography
2	Investigate how cryptographic hashing and digital signatures are used in blockchain. https://101blockchains.com/hashing-and-digital-signature-in-blockchain/ https://coding-bootcamps.com/how-digital-signature-works-in-blockchain/
3	Use tools like Wireshark to analyze cryptographic protocols in real-time network traffic. https://www.wireshark.org/docs/

Suggested Learning Resources:	
Textbooks:	
T1	William Stallings, Cryptography and Network Security - Principles and Practice, Seventh Edition, Pearson Education, 2017.
Reference Books:	
R1	Behrouz A. Ferouzan, Debdeep Mukhopadhyay, Cryptography and Network Security, 3rd Edition, Tata Mc Graw Hill, 2015.
R2	Hans Delfs and Helmut Knebl, Introduction to Cryptography: Principles and Applications, 2nd Edition, Springer Verlag, 2007.
R3	Wenbo Mao, Modern Cryptography, Theory and Practice, 1st edition, Pearson Education, 2003.

Course outcomes: **Upon successful completion of this course, students will be able to**	
CO1	Apply fundamental principles of cryptography to achieve secure communication and protection of information
CO2	Analyze cryptographic mechanisms to identify appropriate solutions for protecting data and communication systems
CO3	Develop secure solutions by integrating suitable cryptographic mechanisms
CO4	Using appropriate computational tools, work ethically and as a team to solve basic security problems

CO PO Mapping

CO/P O	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO1 1	PSO 1	PSO 2	PSO 3
CO1	3													
CO2		3												
CO3			1											
CO4					1		1	1			1			

High-3,Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Information Retrieval	Semester	VI
Course Code	CS372	Course Type	PEC
Total No. of Contact Hours	40	CIA	50
No. of Contact Hours/week (L:T:P)	3:0:0	SEE	50
Credits	3	Total	100
Teaching Department	CS	Exam. Duration	3 hrs

Course Overview

This unit introduces the fundamentals of web search, including how search engines crawl, index, and retrieve web content. It covers web crawling strategies, indexing techniques like inverted indexes, and link analysis algorithms such as PageRank and HITS for ranking web pages based on their link structure

Course objectives: The student will be able to:

- Use the different information retrieval techniques in various application areas
- Apply IR principles to locate relevant information collections of data
- Analyze the performance of retrieval systems when dealing with unmanaged data sources

Prerequisites: None	
Module-1	**8 Hours**
Boolean retrieval. The term vocabulary and postings lists. Dictionaries and tolerant retrieval. Index construction. Index compression. Teaching-Learning Process: Chalk and talk/PPT/case study/web content	
Module-2	**8 Hours**
Scoring, term weighting, and the vector space model. Computing scores in a complete search system. Evaluation in information retrieval. Relevance feedback and query expansion. Teaching-Learning Process: Chalk and talk/PPT/case study/web content	
Module-3	**8 Hours**
XML retrieval. Probabilistic information retrieval. Language models for information retrieval. Text classification. Vector space classification. Teaching-Learning Process: Chalk and talk/PPT/case study/web content	
Module-4	**8 Hours**

Support vector machines and machine learning on documents, Flat clustering, Hierarchical clustering, Matrix decompositions and latent semantic indexing. Teaching-Learning Process: Chalk and talk/PPT/case study/web content

Module-5	**8 Hours**
Web search basics. Web crawling and indexes, Link analysis. Teaching-Learning Process: Chalk and talk/PPT/case study/web content	

Open ended experiments leading to guided projects (Beyond syllabus):

- Develop a Simple Web Crawler: Experiment with crawling specific websites (e.g., news or e-commerce) and storing data locally.
- Build an Inverted Index: Create a keyword-to-document mapping for a small dataset of web pages.
- Simulate Page Rank or HITS Algorithm: Implement and visualize how PageRank changes with different link structures.
- Compare Search Algorithms: Analyse the relevance of results from different query processing or ranking methods.
- Crawl and Analyse Robots.txt Behaviour: Test how different websites handle crawler restrictions.

Suggested Learning Resources: **Textbooks:**	
1	Christopher D. Manning and Prabhakar Raghavan and Hinrich Schütze "Introduction to Information Retrieval" , Cambridge University Press, 2008.
2	Kowalski, Gerald, Mark T Maybury, "Information Storage and Retrieval Systems: Theory and Implementation", Springer.
3	Ricardo Baeza-Yates, "Modern Information Retrieval", Pearson Education, 2007.

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning

- Programming Assignment-1: Implementation of important concepts of data manipulation using NumPy and Pandas (Python)
- Programming Assignment-2: Implementation of simple Machine Learning models with Visualization using Python (MatPlotlib, Scikitlearn)

Course outcomes: **Upon successful completion of this course, students will be able to**	
CO1	Apply fundamental information retrieval concepts and models to solve basic document and web search problems.
CO2	Analyze information retrieval techniques and models to understand their behavior, strengths, and limitations in different retrieval scenarios.
CO3	Evaluate information retrieval and web search methods using appropriate performance measures to determine their effectiveness.
CO4	Collaborate effectively in teams to analyse and evaluate information retrieval problems and present collective solutions.

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3

CO1	3													
CO2		2												2
CO3			1											
CO4								1	1					

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Digital Image Processing	**Semester**	VI
Course Code	CS373	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview:

Digital Image Processing is a comprehensive course that delves into the fundamental principles and practical techniques involved in manipulating and analysing digital images. It establishes a strong foundation in the basics of digital image formation, representation, and the human visual system, progressing to explore essential image enhancement and restoration methods aimed at improving image quality and recovering degraded image.

Course objective is to:

1. Understand the fundamental concepts of image processing.
2. Learn about image formation, image enhancement, feature extraction, and object recognition.
3. Gain practical experience with algorithms and tools related to image processing.

Prerequisite Knowledge: Computer Graphics

Module-1: Image Fundamentals and Enhancement	**8 Hours**

Digital Image Fundamentals: Elements of visual perception, image formation model, sampling and quantization, digital image representation, spatial and intensity resolution.

Intensity Transformations and Spatial Filtering: Intensity transformations (linear, logarithmic, power-law), histogram processing (equalization, matching, specification), spatial filtering basics, smoothing filters (averaging, median, order-statistic, adaptive), sharpening filters (derivative, Laplacian, unsharp masking, high boost, gradient operators).

Activity: Students will perform histogram equalization and apply noise reduction using Python (OpenCV/Matplotlib).

Textbook: Chapter 2, Chapter 3

Module-2: Image Enhancement and Filtering	**8 Hours**

Frequency Domain Processing: Discrete Fourier Transform (DFT), properties of the DFT, frequency domain filtering basics, lowpass filters (ideal, Butterworth, Gaussian), high pass filters (ideal, Butterworth, Gaussian), homomorphic filtering.

Image Restoration: Image degradation/restoration model, noise models, spatial filtering for restoration (mean, order-statistic), frequency domain filtering for restoration (inverse, Wiener).

Activity: problems on Sobel, Prewitt on sample images

Textbook: Chapter 4, Chapter 5

Module-3: Feature Extraction and Image Segmentation	**8 Hours**

Feature Detection: Points, Edges, Corners, Feature Descriptors: SIFT, SURF, ORB, Image Point, line, and edge detection, thresholding (global, adaptive, Otsu's method), region-based segmentation (region growing, splitting and merging), watershed segmentation. **Activity:** Use medical images (Tumour Dataset) to extract features and visualize segmentation results. **Textbook: Chapter 4, Chapter 10**

Module-4: Feature Extraction and Object Recognition	**8 Hours**
Object Recognition: Feature extraction (edges, corners, texture), shape representation (contours, chain codes, shape descriptors), pattern recognition (classifiers like k-NN, SVM), applications (object detection, image classification). **Activity:** Case study on Autonomous vehicle working principle **Textbook: Chapter 11**	

Module-5: Morphological Processing	**8 Hours**
Morphological Image Processing: Dilation and erosion, opening and closing, morphological algorithms (boundary extraction, region filling, thinning, skeletonization). **Activity:** problems on morphological algorithms **Textbook: Chapter 9**	

Suggested Learning Resources: **Textbooks:**	
1	Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing , 4th Edition, Pearson, 2018.
Reference Books:	
1	Dr. Shashidhar Sonnad, Dr. Vybhav K, Dr. P. Joel Josephson, and Dr. Kapil Joshi, Digital Image Processing, 1st edition, Book Rivers, 2022.

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning

Assignments and Quizzes: Regular assignments and quizzes to test understanding of theoretical concepts and practical skills.

Mid-term and Final Exams: Exams covering both theoretical knowledge and practical applications.

Project Work: Comprehensive projects where students apply computer vision techniques to real-world problems.

Class Participation: Active participation in class discussions, group activities, and presentations.

Course outcomes:	
CO1	Apply fundamental concepts of computer vision and image processing.
CO2	Analyse various algorithms for image segmentation and morphological Image processing
CO3	Develop frameworks to apply Machine Learning Algorithms for object recognition
CO4	Work in groups and present their project work.

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3

CO1	3													
CO2		3											3	
CO3			2											
CO4					3			2	2	2				

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Natural Language Processing	**Semester**	VI
Course Code	CS374	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview

This course provides a sound understanding of classical natural language processing. The course provides an insight in to the concepts used and tools available in NLP. While the course does not touch on deep learning based large language models, it provides introduces the concepts of statistical language models

Course objective is to:

1. Give an insight into the fundamentals of natural language processing
2. Provide an understanding of how to extract syntax and semantics of sentences in NLP
3. Introduce NLP techniques for applications like information extraction, information retrieval

Prerequisite Knowledge: Basic understanding of python (needed to do the activities).

Module-1	**Hours: 8**

Overview and language modeling: Overview: Origins and challenges of NLP-Language and Grammar-Processing Indian Languages- NLP Applications-Information Retrieval. Language Modeling: Various Grammar- based Language Models-Statistical Language Model.

Activity: Practice n-gram modelling using functions available in NLTK

Textbook 1: Ch. 1,2

Module-2	**Hours:8**

Word level and syntactic analysis: Word Level Analysis: Regular Expressions-Finite-State Automata-Morphological Parsing-Spelling Error Detection and correction-Words and Word classes-Part-of Speech Tagging. Syntactic Analysis: Context-free Grammar-Constituency- Parsing-Probabilistic Parsing.

Activity: Take a few paragraphs from a text and do POS tagging using NLTK and understand the meaning of various tags used.

Textbook 1: Ch. 3,4

Module-3	**Hours:8**

Semantic Analysis: Introduction: Meaning Representation**:** Lexical Semantics: Ambiguity**:** Word sense disambiguation. **Discourse Processing:** Introduction, Cohesion, Reference Resolution, Discourse Coherence and Structure.

Activity: Install/Use wordnet and understand how to find senses of a few words using it.

Textbook 1: Ch. 5,6

Module-4	Hours:8
NLP Applications: Information Extraction: Named Entity Recognition: Relation Detection and Classification: Temporal and Event Processing: Template-Filling: **Machine Translation:** Why is Machine Translation So Hard? Classical MT & the Vauquois Triangle: Statistical MT, Phrase-based Translation model. **Activity**: Take a text from a newspaper article and perform Named Entity Recognition using NLTK/SpaCy **Textbook 2: Ch. 22.1 to 22.4, 25.1 to 25.4**	
Module-5	**Hours:8**
Information Retrieval And Lexical Resources: Information Retrieval: Design features of Information Retrieval Systems-Classical, Non classical, Alternative Models of Information Retrieval – valuation Lexical Resources: World Net-Frame Net- Stemmers-POS Tagger- Research Corpora. **Activity**: Understand stemming using porter stemming tool lemmatization using WordNet-based lemmatizes. Perform POS tagging using NLTK on a text taken from any textbook. **Textbook 1: Ch. 9,12**	
Follow-on Courses: Deep Learning	

Suggested Learning Resources: **Textbooks:**	
1	Tanveer Siddiqui, U.S. Tiwary, Natural Language Processing and Information Retrieval, First Edition, Oxford University Press, 2008.
2	Daniel Jurafsky, James H. Martin, Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition, 2nd Edition, Pearson Publication, 2014.
Reference Books:	
1	James Allen, Natural Language Understanding, 2nd edition, Benjamin/Cummings publishing company, 1995.
2	Gerald J. Kowalski and Mark.T. Maybury, Information Storage and Retrieval systems, 2nd Edition, Kluwer academic Publishers, 2000.
3	Steven Bird, Ewan Klein and Edward Loper, Natural Language Processing with Python, First Edition, O'Reilly Media, 2009.
Web links and Video Lectures (e-Resources): **NPTEL Course:** https://onlinecourses.nptel.ac.in/noc23_cs45/preview **VTU Resource**: https://online.vtu.ac.in/course-details/Natural-Language-Processing	
Course outcomes: At the end of the course, students will be able to	
CO1	Apply natural language processing techniques for solving real word problems.
CO2	Identify the most suitable methods for specific NLP applications
CO3	Evaluate alternative Models for NLP applications.
CO4	Use natural language processing tools to develop NLP applications.

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	2	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	2	-	-	-	-	-	-	-	-	-

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Introduction to Computer Vision with Deep Learning	**Semester**	VI
Course Code	CS375	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept.	CS	**Course Type**	PEC

Course objectives:

1. Build an intuitive understanding of how computers interpret and learn from images and videos.
2. Learn the fundamental building blocks of deep learning for vision tasks.
3. Gain practical experience using Python and TensorFlow on Google Colab for image classification, detection, and segmentation.
4. Explore real-world and emerging applications of computer vision, including neuromorphic sensing.

PREREQUISITES: Basic Python Programming & Machine Learning Basics.
Familiarity with basic linear algebra and probability concepts is recommended but not mandatory.

Module-1:	**08hours**

Getting Started with Computer Vision and Deep Learning: Introduction to computer vision: what it is, why it matters, and real-world applications. Overview of deep learning and neural networks. Setting up Google Colab and TensorFlow.
Hands-on: Build and train your first image classifier on MNIST digits.
Image Representation and Preprocessing: Pixels, color spaces (RGB, grayscale), resizing, normalization, and data augmentation, and edge detection

Hands-on: Use TensorFlow's Image Data Generator for preprocessing and augmentation.

T1: Ch1.1–1.8, Ch6.1–6.3, Ch9.1–9.4; T2: Ch1.1–1.3, Ch2.1–2.5, Ch3.1–3.7

Module-2:	**08hours**

Understanding Convolutional Neural Networks (CNNs): What are convolutions, filters, feature maps, and pooling? Architecture of a simple CNN.
Hands-on: Build a CNN from scratch for CIFAR-10 image classification.
Improving Model Performance: Overfitting and underfitting. Regularization: dropout, batch normalization, and early stopping.

Hands-on: Tune hyperparameters and visualize learning curves using TensorBoard.

T1: Ch9.1–9.6, Ch7.1–7.8, Ch11.1–11.3; T2: Ch10.1–10.4, Ch12.1–12.3

Module-3:	**08hours**

Transfer Learning and Pre-trained Models: Concept of transfer learning and fine-tuning. Overview of commonly used pre-trained models (VGG, ResNet). Hands-on: Fine-tune a pre-trained model using TensorFlow's Keras Applications.
Object Detection Basics: Difference between classification and detection.. Introduction to bounding boxes and simple detection pipelines.

Hands-on: Implement object detection using TensorFlow's pre-trained models (e.g., SSD or MobileNet).

T1: Ch9 (architectures), Ch11.6–11.7, Ch12.2–12.4; T2: Ch10.4–10.6, Ch14.1–14.5

Module-4:	**08hours**

Semantic Segmentation: What is segmentation and why is it important? Introduction to U-Net (conceptual only).
Hands-on: Perform image segmentation using TensorFlow's ready-to-use APIs.
Model Evaluation and Visualization: Evaluation metrics for classification, detection, and segmentation (accuracy, precision, recall, IoU, F1 Score).
Hands-on: Visualizing predictions using matplotlib and TensorFlow tools. Build a simple dashboard for model evaluation.

T1: Ch9, Ch12.3 + classification metrics; T2: Ch13.1–13.7, Ch14.6–14.7

Module-5:	**08hours**

AI for Real-World Vision Applications: How computer vision powers self-driving cars, healthcare, retail, and security. Responsible AI: bias, fairness, and privacy in visual data. Introduction to model deployment using TensorFlow Lite and web demos.
Hands-on: Convert a trained TensorFlow model to TensorFlow Lite and test inference on Colab.
Introduction to Neuromorphic Vision and Event-Based Imaging: What is neuromorphic vision? Motivation and biological inspiration. Frame-based vs event-based vision: key differences in sensing and data representation.
Event data structures: polarity, timestamp, event streams. Applications: high-speed motion detection, low-power vision for robotics, and autonomous systems.

T1: Ch14, Ch19.1–19.6; T2: Ch1, Ch17.1–17.4, Ch19.1–19.5

Suggested Learning Resources:
Text Books:

T1.	Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press, 2016.
T2.	Richard Szeliski, Computer Vision: Algorithms and Applications, 2nd Edition, Springer, 2022.

Reference Books:

R1.	Adrian Rosebrock, Deep Learning for Computer Vision: Starter Bundle, PyImage Search, 2019.
R2.	Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, New York, 2006.

Course outcomes:

CO1	Apply principles of computer vision, image representation, and foundational deep learning models.
CO2	Analyse Deep Learning Models using TensorFlow/Keras.
CO3	Evaluate, compare, and interpret vision model performance using appropriate metrics.
CO4	Work in group and present their results in research papers

CO-PO Mapping														
CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3													
CO2		3										2		
CO3			2											
CO4					3		3	2	2	2	2			

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Python Programming for Data Science	**Semester**	VI
Course Code	CS376	**Course Type**	PEC
Total No. of Contact Hours	40	**CIA**	50
No. of Contact Hours/week (L:T:P)	3:0:0	**SEE**	50
Credits	3	**Total**	100
Teaching Department	CS	**Exam. Duration**	3 hrs

Course Overview: This course introduces students to Python programming fundamentals and its powerful data handling libraries—NumPy and Pandas. Structured over five modules, the course begins with core language constructs and progresses into structured programming and data analysis tools. Through practical examples and exercises, students will develop the skills necessary for real-world programming and data manipulation using Python.

Course objectives: The student will be able to:

- To understand Python constructs and use them to build the programs
- To analyse different conditional statements and their applications in programs.
- To learn and use basic data structures in python language.
- To learn and demonstrate array manipulations by reading data from files
- To understand and use different data in a data analytics context.

Prerequisites: Programming Skills

Module-1	**8 Hours**
Introduction to python: Elements of python language, python block structure, variables and assignment statement, data types in python, operations, simple input/output print statements, formatting print statement. Text Book 1: Chapter 3 (3.2, 3.3, 3.4, 3.6, 3.7, 3.9 and 3.10)	
Module-2	**8 Hours**
Decision structure: forming conditions, if statement, the if-else and nested if-else, looping statements: introduction to looping, python built in functions for looping, loop statements, jump statement. Text Book 1: Chapter 4 (4.2 to 4.6) , Chapter 5 (5.1 to 5.4)	
Module-3	**8 Hours**
Lists: lists, operation on list, Tuples: introduction, creating,indexing and slicing, operations on tuples. sets: creating, operation in sets, introduction dictionaries, creating, operations, nested dictionary, looping over dictionary. Text Book 1: Chapter 7 (7.2 to 7.3) , Chapter 8 (8.1 to 8.4) and Chapter 9(9.1 to 9.3, 9.7 to 9.12)	

Module-4	**8 Hours**
The NumPy Library: Ndarray: the heart of the library, Basic operations, indexing, slicing and iterating, conditions and boolean arrays, array manipulation, general concepts, reading and writing array data on files. The pandas Library: an introduction to Data structure, other functionalities on indexes, operations between data structures, function application and mapping. Text Book 2: Chapter 3 and Chapter 4.	
Module-5	**8 Hours**
The pandas: Reading and Writing data: i/o API tools, CSV and textual files, Reading data in CSV or text files, reading and writing HTML files, reading data from XML files, Microsoft excel files, JSON data, Pickle python object serialization. Pandas in Depth : data manipulation: data preparation, concatenating data transformation discretization binning, permutation, string manipulation, data aggregation group iteration. Text Book 2: Chapter 5 and Chapter 6	

Suggested Learning Resources: **Textbooks:**	
1	S. Sridhar, J. Indumathi, V.M. Hariharan, Python Programming, Pearson publishers, 1st edition 2023.
2	Fabio Nelli, Python Data Analytics,1st Edition, , Apress, Publishing, 2015.
Reference Books:	
1	Paul Deitel and Harvey Deitel, Intro to Python for Computer Science and Data science, 1st edition Pearson Publisher 2020.

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning

- Assessment Methods
- Programming Assignment

Course outcomes: At the end of the course, the student will be able to	
CO1	Apply core Python programming concepts to simple Python programs.
CO2	Analyze and implement Python programs to manipulate, process, and interpret structured datasets.
CO3	Evaluate data handling and transformation techniques in Python to select appropriate methods for real-world data processing tasks.
CO4	Work collaboratively in teams to analyze data-oriented problems using Python libraries and present the solutions effectively.

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3													
CO2		2										2		
CO3			1											
CO4								2	2					

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Social Network Analysis	**Semester**	VI
Course Code	CS377	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview

This course provides an in-depth exploration of social networks, their structures, and their impact on society. Students will gain knowledge of how social networks function, how human behavior influences network dynamics, and how to apply analytical and visualization techniques for social network analysis.

Course objective is to:

1. Explore the Fundamentals of Social Networks and Their Applications
2. Analyze Human Behavior in Social Web and Communities
3. Develop Analytical and Visualization Skills for Social Network Analysis

Prerequisite Knowledge: Basic Mathematics & Statistics, Programming Knowledge, Data Science & Analytics Basics,Introduction to Social Media & Web Technologies, Critical Thinking & Problem-Solving Skills

Module-1	**08 Hours**

Introduction: Applications, Preliminaries, Three Levels of SNA, Historical Development, Graph Visualization Tools.

Network Measures: Network Basics, Node Centrality, Assortativity, Transitive and Reciprocity, Similarity.

Activity: Node Centrality Exercise: Students calculate simple centrality measures for the given graph, MCQ's

Textbook 1: Chapter 1, 2

Module-2	**08 Hours**

Network Growth Models: Properties of real-world networks, Random network model, Ring Lattice Network Model, Watts Strogatz model. Preferential Attachment model, Price's model.

Link Analysis: Applications, Signed networks, Strong and weak Ties, Link analysis and algorithms, Page Rank, Personalized Page Rank, Divrank, Simrank.

Activity: PageRank Exploration:Students manually calculate a simplified PageRank for a small network, Practical implementation using scikit-learn, Students work in groups to map their social networks and classify ties as strong or weak.

Module-3	**08 Hours**

Community Structure In Networks: Applications, Types of communities, Community detection methods, Disjoint community detection, overlapping community detection, local community detection, community detection vs community search.

Link Prediction: Applications, temporal changes in a network, Problem definition, evaluating link prediction networks, Heuristic Models, probabilistic Models.

Activity: Link Prediction Exercise: provides a partially completed network graph where some links are missing & make the students fill it, Practical implementation using scikit-learn.
Textbook 1: Chapter 5.1 to 5.7, 6.1 to 6.6.

Module-4	**08 Hours**
Cascade Behavior And Network Effects: Preliminaries, Cascade model, case study, probabilistic cascades, epidemic models, independent cascade models, Cascade prediction. **Anomaly Detection In Static Networks:** Outliers vs Network-based anomalies, challenges, Anomaly detection in static and dynamic networks. **Activity**: Case Study Analysis, Detecting Network Anomalies from the given graph & justify the answer. **Textbook 1: Chapter 7, 8**	
Module-5	**08 Hours**
Graph Representation Learning: Machine learning pipelines, Intuition behind representation learning, benefits, criteria of GRL, GRL pipelines. **Application And Case Studies**: Malicious activities on OSNs, Sock puppets in OSNs, Modeling the spread of COVID 19, Recommendation System. **Activity**: Recommendation System Brainstorm, Quiz **Textbook 1: Chapter 9.1 to 9.5, 10.1,10.2,10.4,10.5**	
Follow on courses: Advanced Social Network Analysis, Machine Learning for Social Media Analytics, Cybersecurity and Privacy in Social Networks	

Suggested Learning Resources: **Textbooks:**	
1.	Tanmoy Chakraborty "Social Network Analysis" 1st edition, Wiley, 2021.
Reference Books:	
1	Albert-Lazzlo Barabasi, Network Science, 1st edition, Cambridge university Press, 2016.
2	Stanley Wasserman, Katherine Faus, Social Network Analysis: Methods and Applications, first edition, 2019.
Web links and Video Lectures (e-Resources): **M1: https://onlinecourses.nptel.ac.in/noc22_cs117/preview**	
Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning Quiz, Group Discussions, Seminars, In class assignments.	

Course outcomes:	
CO1	Apply fundamental concepts and analytical techniques to solve problems in social networked systems.
CO2	Analyze network data using appropriate techniques to identify key patterns and insights.
CO3	Design analytical models and workflows using social network measures and algorithms to address real-world network-based problems.
CO4	Use appropriate tools to analyze and visualize data and communicate insights effectively in a collaborative environment.

CO-PO Mapping

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	2	-
CO3	-	-	2	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	3	-	-	1	1	-	1	-	-	-

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	System Modelling and Simulation	**Semester**	VI
Course Code	CS378	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview

The System Modeling and Simulation course provides a comprehensive understanding of how to analyze, design, and evaluate complex systems using simulation techniques. It covers the fundamental principles of system modeling, statistical analysis, queuing theory, and simulation methodologies, equipping students with the skills necessary to model real-world scenarios effectively.

Course objectives

- To introduce students with Fundamentals of Simulation and develop Simulation Models of real-world systems using discrete-event simulation techniques and implement using spreadsheets, Java, and GPSS.
- Students will be able to apply Statistical Models in Simulation, Generate and Validate Simulation Data, Perform Output Analysis & Optimization, Validate and Calibrate Simulation Models

Prerequisite Knowledge: Students should have prior knowledge in the following: Mathematics and Statistics, Basic Programming skills.

Module-1: Introduction to Simulation **8 Hours**

When simulation is the appropriate tool and when it is not appropriate; Advantages and disadvantages of Simulation; Areas of application; Systems and system environment; Components of a system; Discrete and continuous systems; Model of a system; Types of Models; Discrete-Event System Simulation; Steps in a Simulation Study. The basics of Spreadsheet simulation, Simulation example: Simulation of queuing systems in a spreadsheet.

Activity:

1. Analyze a given system (e.g., ATM, traffic system, hospital emergency room) and classify its components (discrete/continuous, input/output, etc.)
2. Implement a simple queuing system (e.g., a single-server queue) using Microsoft Excel or Google Sheets.

Textbook Chapters: chapter 1 & chapter 2

Module-2: General Principles, SimulationSoftware & Statistical Models in Simulation **8 Hours**

Concepts in Discrete-Event Simulation: The Event-Scheduling / Time-Advance Algorithm, World Views, Manual simulation Using Event Scheduling; List processing. Simulation in Java; Simulation in GPSS. Review of terminology and concepts used in statistical modelling; Useful statistical models; Discrete distributions; Continuous distributions; Poisson process; Empirical distributions.

Activity:

1. Implement a basic event-scheduling algorithm in Python/Java

2. Match given real-world datasets to appropriate statistical distributions (e.g., normal, exponential, Poisson).

Textbook Chapters: chapter 3, chapter 4 & chapter 5

Module-3: Queuing Models, Random-Number Generation	**8 Hours**

Characteristics of queuing systems; Queuing notation; Long-run measures of performance of queuing systems; Steady-state behaviour of M/G/1 queue; Networks of queues; Rough-cut modelling: An illustration. Properties of random numbers; Generation of pseudo-random numbers; Techniques for generating random numbers; Tests for Random Numbers

Activity:

1. Model and analyze different queuing scenarios (e.g., M/M/1, M/G/1) using simulation tools like SimPy (Python)
2. Write a program to generate pseudo-random numbers and validate them using statistical tests.
3. Analyze research papers on queuing theory applications in real-world problems.

Textbook Chapters: chapter 6 & chapter 7

Module-4: Input Modeling & Random-Variate Generation	**8 Hours**

Data Collection; Identifying the distribution with data; Parameter estimation; Goodness of Fit Tests; Fitting a non-stationary Poisson process; Selecting input models without data; Multivariate and Time-Series input models. Random-Variate Generation: Inverse transform technique; Acceptance-Rejection technique; Special properties.

Activity:

1. Collect real-world data (e.g., arrival times at a store) and determine an appropriate probability distribution.
2. Apply statistical tests (Chi-Square, KS test) to validate the fitted distribution.
3. Implement inverse transform and acceptance-rejection techniques in Python.

Textbook Chapters: chapter 8 & chapter 9

Module-5: Output Analysis, Verification and Optimization	**8 Hours**

Types of simulations with respect to output analysis; Stochastic nature of output data; Absolute measures of performance and their estimation; Output analysis for terminating simulations; Output analysis for steady-state simulations. Model building, verification and validation; Verification of simulation models; Calibration and validation of models, Optimization via Simulation.

Activity:

- Implement a Monte Carlo simulation for a decision-making problem (e.g., stock price prediction, risk analysis).
- Conduct output analysis for a simulated system and compare terminating vs. steady-state results.
- Given a dataset, validate a simulation model by comparing simulated vs. real-world data.

Textbook Chapters: chapter 10 ,chapter 11 & chapter 12

Follow-on Courses: Advanced Model-Based System Design and ML-Aided Simulation.

Suggested Learning Resources:

Textbook:

1	Jerry Banks, John S. Carson II, Barry L. Nelson, David M. Nicol: Discrete-Event System Simulation, 5th Edition, Pearson Education, 2010.

Reference Books:

1	Lawrence M. Leemis, Stephen K. Park, Discrete – Event Simulation, A First Course, 1st Edition, Pearson Education, 2006.

2	Averill M. Law: Simulation Modeling and Analysis, 4th Edition, Tata McGraw-Hill, 2007.

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning

Case Study Analysis:

- **Team-Based Simulation Project:** Build and validate a complete simulation model (e.g., airport security checkpoint, hospital emergency room, or traffic system).

Course outcomes:	
CO1	Apply principles of simulation, statistical modeling, and computational tools to model real-world systems.
CO2	Analyze simulation models to identify system behavior, underlying assumptions, and performance measures.
CO3	Develop simulation models using appropriate techniques and tools to solve system-level problems and decision-making scenarios.
CO4	Carry out simulation activities using statistical techniques, random-number generation, and simulation tools to model simple real-world systems.

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	1	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	1	1	-	-	-	-	-

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Advanced Java	**Semester**	VI
Course Code	CS379	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week(L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview

This course covers essential concepts for building robust and scalable Java applications. It covers Generics and Collections for efficient data handling, String Handling for manipulation techniques and JDBC Connectivity for database integration. It explores Servlets, JSP, Hibernate and Spring Framework for dependency injection, MVC and enterprise application development.

Course objectives:

1. Apply the concepts of Generic and Collections Framework
2. Demonstrate the fundamental concepts of String operations
3. Apply database interaction through Java database Connectivity
4. Design and develop web applications using Java servlets and JSP
5. Using Hibernate Framework to interact with the database using OOP.

Prerequisite Knowledge: This course requires that the students are familiar with programming language such as Java.

Module-1: String Handling	**8 Hours**

The String Constructors, String Length, Special String Operations, Character Extraction, String Comparison, Searching Strings, Modifying a String, Data Conversion Using valueOf( ), Changing the Case of Characters Within a String, joining strings, Additional String Methods, StringBuffer, StringBuilder

Activity: Implement a string-based password validator

Text Book 1: Ch 18

Module-2: Generics and Collections Framework	**8 Hours**

Generics: What are Generics, A Simple Generics Example. Collections Framework: Collections Overview, **The Collection Interface:** List, Set, SortedSet, The Collection Class: ArrayList, LinkedList, HashSet, LinkedHashSet, TreeSet, Accessing a collection via an Iterator.

Activity: Iterator Scavenger Hunt

Textbook 1: Chapter 14

Module-3: JDBC Connectivity	**8 Hours**

The concept of JDBC; JDBC Driver Types; JDBC packages; A brief overview of the JDBC Process; Database Connection; Associating the JDBC/ODBC Bridge with the Database; Statement Objects; ResultSet; Transaction Processing; Metadata, Data Types; Exceptions.

Activity: Implement a basic login system that verifies users from a database

Textbook 2: Chapter 6

Module-4: Servlets and JSP	**8 Hours**
Background; The life cycle of a servlet; A simple servlet; the servlet API; The javax.servlet package Reading servlet parameter; the javax.servlet.http package; Handling HTTP Requests and Responses; using Cookies; Session Tracking, Java Server Pages (JSP); JSP tags, Variables and Objects, Methods, Control statements, Loops, Request String, Parsing other information, User sessions, Cookies, Session Objects. **Activity:** Students review each other's session-based applications and suggest improvements. **Textbook 1: Chapter 36** **Textbook 2: Chapter 11**	
Module-5: Junit and Hibernate Framework	**8 Hours**
JUnit for Unit testing Java applications, Developing a simple Maven Project. Hibernate Framework: Overview, Architecture, Configuration, Sessions, Mapping Files, Query Language, Simple Hibernate Application. **Activity:** Create a Hibernate-based CRUD (Create, Read, Update, Delete) application. **Case Studies: cloud-Based File Sharing, Payment Gateway Integration**	
Follow-on courses: Android Application Development	

Suggested Learning Resources: **Textbooks:**	
1.	Herbert Schildt, JAVA the Complete Reference, 13th Edition, Tata McGraw-Hill, 2024.
2.	Jim Keogh, The Complete Reference J2EE, 1st edition, Tata McGraw-Hill, 2017.
Reference Books:	
1.	Y. Daniel Liang: Introduction to JAVA Programming, 7th Edition, Pearson Education, 2007.
Web links and Video Lectures (e-Resources): 1. https://nptel.ac.in/courses/106/105	
Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning 1. Programming exercises	

Course outcomes:	
CO1	Apply advanced Java programming concepts to implement structured and modular software applications.
CO2	Analyze programming problems to obtain optimal solutions using object-oriented features.
CO3	Develop maintainable and robust Java applications using advanced programming constructs.
CO4	Use Java tools and frameworks to implement, test, and debug applications effectively.
CO5	Demonstrate the ability to work independently and collaboratively in a team to design, implement, and solve programming tasks in Java.

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3													
CO2		3												
CO3			2									2		
CO4				1	3									
CO5							1	1	1		1			

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Cloud Computing	**Semester**	VII
Course Code	CS401	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course overview:

Cloud Computing is a modern computing model that delivers on-demand computing resources—such as servers, storage, databases, networking, and software—over the internet. It eliminates the need for organizations to own and maintain physical infrastructure, allowing them to scale resources as needed efficiently. It is widely adopted across industries like healthcare, finance, and education, playing a crucial role in supporting modern applications such as AI, IoT, and big data analytics.

Course objective is to:

1. Summarize the importance of cloud computing in the real world.
2. Identify applications that can be integrated using cloud services.
3. Evaluate cloud-based applications.
4. Analyze the security issues in cloud services.
5. Classify the Cloud Services and Cloud related technologies.

Prerequisites: Computer Network, Operating System and Database Management System

Module-1: Introduction to Cloud Computing	**08 Hours**

Introduction: Evolution and History of Cloud Computing, Introduction to Cloud Computing, Why Cloud Computing is Becoming Highly Important? Cloud Types, Layers and Types of Cloud, Features of Cloud Computing, Why Cloud Computing? Cloud Computing Stack, Component of Cloud Computing, Advantages of Cloud Computing, Limitations of Cloud Computing, Cloud Computing Service Providers

Virtualization: Introduction to Virtualization and its Technical Evolution, History of Virtualization, Introduction to Virtualization, From Virtual Computing to Clouds, Load Balancing and Virtualization, Virtualization Security Management

Case study: Cloud Adoption and Virtualization on any University

Textbook 1: Chapter:1,2

Module-2: PaaS and SaaS	**08 Hours**

SOFTWARE AS A SERVICE: Evolution of SaaS, Brief Introductory part of Software as a Service, SaaS Unification Technologies, SaaS Integration Products and Technologies, SaaS Product Selection Criteria, SaaS Integration Services, Advantages of SaaS.

PLATFORM AS A SERVICE: Evolution of PaaS, Introduction to PaaS, PaaS Service Providers, PaaS Application Framework, PaaS Operator Verbs, PaaS Developer Verbs, Advantages and Challenges of PaaS

Case study: Implementing SaaS and PaaS at LearnSmart Institute

Textbook 1: Chapter :3-4

Module-3: Infrastructure as a Service and Data in Cloud	**08 Hours**

Infrastructure as a Service : Evolution, IaaS Architecture, IaaS Providers, Experimental Setup, Advantages and Disadvantages of Infrastructure as a Service.
Data in Cloud: Evolution of Network Storage in Cloud, Data as a Service, Database as a Service, Cloud Based Data Storage, Advantages and Limitations of Cloud Based Storage Solution, Cloud Based Data Storage Service Providers, Cloud Storage Interoperability. **Case study**: Migrating University Infrastructure and Data to Cloud – A Case of BrightFuture University **Textbook 1:** Chapter 5-6

Module-4 : Cloud Security and Simulation	**08 Hours**
Cloud Security: Introduction, Security Risks and Best Practices, Securing Cloud, Security Policy Implementation, Techniques to Secure Data, Cloud Encryption, Cloud Security Alliance, Cloud Security Strategy, Cloud Computing Security Architecture, Future of Cloud Security **Cloud Computing – Simulation:** Introduction to Simulation, Cloud Computing: Simulation Tools, Cloud based Web Applications & Service Testing Tools, Cloud Based Mobile & Multimedia Application Testing Tools, Performance Comparison of Cloud Computing Simulators. **Case study**: Ensuring Cloud Security and Performance Testing Using Simulation – A Case of TechNova College **Textbook 1**: Chapter 7 and 8	

Module-5: Resource Allocation in Cloud Computing and Mobile Cloud Computing	**08 Hours**
Introduction, Importance of Resource Allocation, Strategies for Resource Allocation, Resource Allocation Policies and Algorithms. **Mobile Cloud Computing:** Introduction to Mobile Cloud Computing, Features of Mobile Cloud Computing, Mobile Cloud Computing Service Models, Various Architectures of Mobile Cloud Computing, Issues Surrounding Mobile Cloud Computing. **Case study**: Optimizing Resource Allocation and Mobile Cloud Services for SmartCampus App at GlobalTech University **Textbook 1:**Chapter 10	

Suggested Learning Resources: **Textbooks:**	
1	Dr. Anand Nayyar, Handbook of Cloud Computing , First Edition, BPB publications, 2019.
Reference Books:	
1	Arshdeep Bahga, Vijay Madisetti, Cloud Computing – A Hands-on Approach, 1st Edition, Universities press pvt Ltd, 2016.
2	Barrie Sosinsky, Cloud Computing Bible, 1st Edition, Wiley India Pvt Ltd, 2011.
3	Robert Elsenpeter, Toby J. Velte, Anthony T. Velte, Cloud Computing: A Practical Approach, 1st Edition, TataMcGrawHill Education, 2011.
Web links and Video Lectures (e-Resources): a) https://nptel.ac.in/courses/106/105/106105167/	

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning
Assignments and Quizzes: Regular assignments and quizzes to test understanding of theoretical concepts and practical skills. **Project Work:** Comprehensive projects where students work with AWS, NEBULA etc **Class Participation:** Active participation in class discussions, group activities, and presentations.

Course outcomes:	
CO1	Understand and articulate the significance of cloud computing in modern technology-driven environments.
CO2	Identify and design cloud-based solutions for real-world applications.
CO3	Evaluate the performance and efficiency of cloud applications in various scenarios.
CO4	Analyze and address security threats in cloud computing services and recommend solutions.
CO5	Explore how to manage cloud.

CO-PO Mapping

CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	-	1	-	-	-	-	-	-	-	-	3	2	3
CO2	2	-	2	2	1	-	-	-	-	-	-	3	2	1
CO3	2	-	3	2	1	-	-	-	-	-	-	3	2	2
CO4	2	-	3	3	2	1	-	-	-	-	-	3	2	2
CO5	2	-	3	3	2	2	-	-	-	-	-	3	3	1

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Cloud Computing Laboratory	**Semester**	VII
Course Code	CS402	**CIA**	50
L:T:P	0:0:2	**SEE Marks**	50
Hrs. / Week	2	**Total Marks**	100
Credits	1	**Exam Duration**	03 Hours
Teaching Department	CS		

Course objectives: This laboratory course enables students to	
1.	Fundamental and practical knowledge of cloud computing and the OpenStack platform.
2.	Students will learn to install, configure, and manage OpenStack services, automate resource provisioning, and implement advanced features like identity management, load balancing, and autoscaling.
3.	Hands-on experience in deploying scalable cloud solutions for real-world applications.

Pgm. No.	**List of Experiments / Programs**	**Hour s**	**Cos**
Prerequisite Knowledge: This course requires that the students are familiar Understanding of IP addressing, subnets, VLANs, and basic network troubleshooting.			
	PART-A		
1	**Title**: Install Virtualbox / VMware Workstation with different flavours of linux or windows OS on top of windows7 or 8. **Problem Description**: create a virtual environment on a Windows 7 or 8 host operating system where you can run multiple operating systems (e.g., Linux distributions like Ubuntu, Fedora, CentOS, or another version of Windows). This allows testing, development, or learning without modifying the host OS. **Method**: a) System Requirements b) Download Virtualization Software c) Install Virtualization Software d) Enable Virtualization in BIOS/UEFI e) Download the OS ISO Files f) Create a Virtual Machine g) Install the Guest OS h) Configure Networking and Shared Folders (Optional) **i)** Test and Use the Virtual Machines **Theory Reference: Module 1**	2	1, 2
2	**Title:** Install a C compiler in the virtual machine created using virtual box and execute Simple Programs.	2	1, 2

	Problem Description: Install a C compiler in a virtual machine created using VirtualBox to compile and execute simple C programs. The virtual machine can run any operating system (Linux or Windows). This setup allows you to write, compile, and run C programs in an isolated environment. **Method:** 1. Install C Compiler on Linux OS. **2.** Install C Compiler on Window OS **Theory Reference: Module 1**		
3	**Title:** Install Google App Engine. Create hello world app and other simple web applications using python/java. **Problem Description:** Install **Google App Engine (GAE)** on your system, create a "Hello, World!" app, and develop simple web applications using Python or Java. GAE provides a platform to build, test, and deploy web applications on Google Cloud. **Method:** d) Install Google Plugin for Eclipse. e) Create New Web Application Project f) Deploy to Google App Engine **Theory Reference: Module 2**	2	1, 2
4	**Title:** Install Hadoop single node cluster and run simple applications like wordcount. **Problem Description:** The task is to install Hadoop in a single-node cluster setup and run a simple WordCount application. This involves setting up Hadoop on a single machine (acting as both the master and slave node), configuring the Hadoop environment, and executing a basic MapReduce job (WordCount) to process text data. **Method:** 3. Install Hadoop 4. Configure Hadoop 5. Format the HDFS filesystem 6. Start Hadoop services 7. Run the WordCount MapReduce job **8.** Check the output **Theory Reference: Module 2**	2	1, 2
5	**Title:** Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim. **Problem Description:** CloudSim is a framework for modeling and simulating cloud computing environments, including data centers, virtual machines, and cloud services. It provides support for testing	2	1, 2

	and evaluating scheduling algorithms and resource allocation policies in a simulated cloud environment. **Method:** a) Set Up CloudSim b) Design a Custom Scheduling Algorithm c) Modify CloudSim Classes d) Create the Simulation Environment e) Run the Simulation **Theory Reference: Module 3**		
6	**Title:** Find a procedure to launch virtual machine using trystack (Online Openstack Demo Version) **Problem Description:** The task is to launch a Virtual Machine (VM) using Trystack, which is an online OpenStack demo version that allows users to interact with OpenStack cloud infrastructure without needing to set up their own OpenStack environment. **Method:** a) Sign up for Trystack and log in to the dashboard. b) Create a Key Pair for SSH access to the VM. c) Create a Security Group to allow necessary ports (SSH, HTTP). d) Launch a Virtual Machine by selecting the image, flavor, key pair, and security group. e) Once the VM is running, access it via SSH using the private key. f) Manage the VM (start/stop/reboot/delete) via the Trystack Dashboard **Theory Reference: Module 4**	2	1, 2
7	**Title:** Use GAE launcher to launch the web applications. **Problem Description: GAE Launcher** or an equivalent tool to locally test and then deploy simple web applications, such as a "Hello, World!" app or more complex applications, using Python or Java. This involves configuring the app environment, running it locally for testing, and deploying it to Google Cloud. **Method:** 1. Install and set up the required tools (GAE Launcher or Cloud SDK) to run and deploy web applications. 2. Write a simple web application (e.g., "Hello, World!") using Python or Java. 3. Test the application locally using a development server. 4. Deploy the application to Google App Engine. 5. Access the application online via the provided URL.	2	4, 5

	Theory Reference: Module 5		
8	**Title: Deploying an App using Google App Engine** **Problem Description:** Students will build a basic web application using Python or Node.js and deploy it using Google App Engine, experiencing the full application lifecycle and learning about App Engine's architecture and cost model. **Method:** a) Set Up Environment b) Develop Application c) Deploy the App d) Understand Cost Model e) Observations **Theory Reference: Module 5**	2	4,5

PART-B		
Mini-Project	**Mini-Project:** 1. Cloud-Based File Storage and Sharing System 2. Cloud Resource Management System 3. Cloud-Based Chat Application 4. Cloud-Based Task Scheduler 5. Cloud-Based Healthcare Record Management System 6. Cloud Cost Optimization Tool 7. Cloud-Based IoT Device Management 8. Serverless E-Commerce Application 9. Disaster Recovery Simulation in Cloud 10. Cloud-Based Video Processing	3

Beyond Syllabus Virtual Lab Content

1. Perform live migration of a virtual machine using VMware or OpenStack.
2. Experiment with workload balancing in a multi-cloud setup.
3. Create a disaster recovery plan and simulate data restoration using cloud backup service

Suggested Learning Resources:
Textbooks:

1	Arshdeep Bahga and Vijay Madisetti "Cloud Computing: A Hands-On Approach", 1st Edition, self-published by the authors,2014.
2	Michael J. Kavis "Architecting the Cloud: Design Decisions for Cloud Computing Service Models (SaaS, PaaS, and IaaS)" 1st edition, Wiley, 2014.

Reference Books:

1	Mr. J. K. J. Singh "Cloud Computing: From Beginning to End"

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning

Implementing Cloud-Based IoT (Internet of Things) Solution.

Deploying a Cloud-Based Database

Course outcomes: At the end of the course, students should be able to	
CO1	Outline Cloud Computing Fundamentals
CO2	Apply Cloud Platforms and Tools
CO3	Create Serverless Computing and Microservices
CO4	Plan Cloud Security and Identity
CO5	Design and Deploy Cloud-Based Applications

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	-	-	-	3	2	-	-	-	-	-	-	3	2
CO3	-	-	3	-	3	2	-	-	-	-	2	-	2	3
CO4	-	-	-	-	-	2	2	3	-	-	-	-	-	3
CO5	-	-	3	2	3	-	-	-	-	-	3	-	3	3

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Advanced Databases & Analytics	**Semester**	VII
Course Code	CS403	**CIA**	50
Total No. of Contact Hours	(40 Th+20 Pr)=60Hrs	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:2	**Total**	100
Credits	4	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview: This course provides an in-depth understanding of modern database systems, from traditional relational models to contemporary big data and NoSQL technologies. With a blend of theory and hands-on activities, students will explore advanced topics including object-extensions, distributed databases, NoSQL, data warehousing, and data visualizations.

Course objective is to:

- Analyze database system architecture and performance optimization techniques.
- Explore advanced database models (e.g., object-oriented, NoSQL, graph databases).
- Study distributed database systems and data replication strategies.
- Understand data warehousing and data mining concepts.

Prerequisite Knowledge: This course requires that the students are familiar with basic database concepts and SQL.

Module-1: Relational and Object-Relational Databases **8 Hours**

Review of Relational Data Model and Relational Database Constraints: Relational model concepts; Relational model constraints and relational database schemas; Update operations, anomalies, dealing with constraint violations, Types and violations.

Object and Object-Relational Databases: Overview of Object Database Concepts, Object Database Extensions to SQL, The ODMG Object Model and the Object Definition Language ODL, Object Database Conceptual Design, The Object Query Language OQL.

Activity: Consider the relations for a database that keeps track of student enrollment in courses and the books adopted for each course. Identify the primary and foreign keys, illustrate constraint violations.

Text Book 1: 5.1, 5.2, 5.3, 12.1 to 12.5

Module-2: Disk Storage and Distributed Databases **8 Hours**

Disk Storage, Basic File Structures: Introduction, Secondary Storage Devices, Buffering of Blocks, Placing File Records on Disk Operations on Files, Files of Unordered Records (Heap Files), Files of Ordered Records (Sorted Files).

Distributed Database Concepts: Distributed Database Concepts, Data Fragmentation, Replication, and Allocation Techniques for Distributed Database Design, Overview of Concurrency Control and Recovery in Distributed Databases, Overview of Transaction Management in Distributed Databases, Query Processing and Optimization in Distributed Databases, Types of Distributed Database Systems, Distributed Database Architectures, Distributed Catalogue Management.

Activity: Designing a Distributed Database for a Global E-commerce Company.
Text Book 1: 16.1 to 16.8, 23.1 to 23.4, 23.7

Module-3: NoSQL Databases **8 Hours**

NoSQL Databases and Big Data Storage Systems: Introduction to NoSQL Systems, The CAP Theorem.

Big Data Technologies Based on MapReduce and Hadoop: What Is Big Data? Introduction to MapReduce and Hadoop, Hadoop Distributed File System (HDFS), MapReduce: Additional Details Hadoop v2 alias YARN.

Activity: Designing a NoSQL Database for a Social Media Platform.

Text book 1:24.1,24.2, 25.1 to 25.5

Module-4: Enhanced Data Models	**8 Hours**

Introduction to Active, Temporal, Spatial, Multimedia, and Deductive Databases: Active Database Concepts and Triggers, Temporal Database Concepts, Spatial Database Concepts, Multimedia Database Concepts, Introduction to Deductive Databases.

Introduction to Information Retrieval and Web Search: Information Retrieval (IR) Concepts, Retrieval Models, Types of Queries in IR Systems, Text pre-processing, Inverted Indexing, web Search and Analysis.

Activity: Designing an Active Database by adding trigger for Inventory Management.

Text book 1:26.1 to 26.5, 27.1 to 27.5, 27.7

Module-5: Data Warehousing & Visualizations	**8 Hours**

Overview of Data Warehousing: Overview of Data Mining Technology, Applications of Data Mining.

Visualizations (TABLEAU): Introduction to Data Visualization, Tableau Interface Overview, Connecting to Data. **Building Basic Visualizations & Dashboards**: Bar charts, Line charts, Scatter plots, Advanced Chart Types, Advanced Visualizations & Data.

Activity: Analyze sample dashboards related to engineering data.

Text book1: 28.1, 28.6, Text book2

Follow on Courses: Data Warehousing and Business Intelligence

Resources for Advanced Learners:
Database Systems: The Complete Book by Hector Garcia-Molina, Jeffrey D. [1] Ullman, and Jennifer Widom.

Practice Component of IPCC	**L:T:P (0:0:2)**

Programming Experiments (Suggested and are not limited to)

Program 1:
Title: Library Database Design and Implementation in SQL.

Problem Description: Writing SQL queries to perform the following operations: Insert a new book and author, borrow a book by a member, return a book, find all books by a specific author, list all members who have borrowed a specific book.

Method: Design a relational database schema to represent the following entities and their relationships:
Book: ISBN, Title, Publication Year, Publisher, Genre
Member: MemberID, Name, Address, Phone Number Author:
AuthorID, Name, Birth Year
Loan: LoanID, ISBN, MemberID, Loan Date, Return Date.

Theory Reference: Module 1

Program 2:
Title: Develop a database application to demonstrate storing and retrieving of BLOB and CLOB objects.

Problem Description: Write a binary large object (BLOB) to a database as either binary or character (CLOB) data, depending on the type of the field in your data source. To write a BLOB value to the database, issue the appropriate INSERT or UPDATE statement and pass the BLOB value as an input parameter. If your BLOB is stored as text, such as a SQL Server text field, pass the BLOB as a string parameter. If the BLOB is stored in binary format, such as a SQL Server image field, pass an array of type byte as a binary parameter.

Method: Parameterized queries (or prepared statements) with type handling.
Theory Reference: Module 1

Program 3:
Title: Develop a database application to demonstrate the representation of multi valued attributes, and the use of nested tables to represent complex objects. Write suitable queries to demonstrate their use.

Problem Description: Implementing the Application under the Relational Model using only Oracle's built-in data types. Build an object-oriented application on top of this relational schema using object views.

Method: Using nested tables (or similar object-relational features) to represent complex objects and multi-valued attributes within a relational database.

Theory Reference: Module 1

Program 4:
Title: Simulating File Organization.

Problem Description: Develop a simple file management system and implement different file organization methods and compare their performance for various operations.

Method: Create a program (e.g., in Python) that simulates a heap file. Data records should be simple structures (e.g., containing an ID and a name). Implement functions to: Insert a new record, search for a record by ID (linear search), delete a record by ID, display all records.
Theory Reference: Module 2

Program 5:
Title: Analyzing Movie Data with MongoDB and MapReduce.

Problem Description: Analyze a large movie dataset (titles, genres, ratings, cast) using MongoDB queries.

Method: This lab program combines NoSQL database concepts using MongoDB with Big Data processing using MapReduce (simulated or using a simplified framework).

Theory Reference: Module 3

Program 6:
Title: Design a Student Database application, including attendance records for each enrolled subject and internal assessment marks. Implement database triggers to enforce data integrity and automate updates related to attendance and internal assessment calculations.
Problem Description: Database triggers for data integrity and automation and to implement an active database.

Method: Using Database triggers with a Student Database application.
Theory Reference: Module 4

Program 7:
Title: Design a Web Search Simulation and perform Analysis.

Problem Description: Implement a simple web crawler that starts from a few seed URLs and follows links to other pages.

Method: A web crawler (you can use existing libraries like Scrapy in Python or build a very basic one for this experiment's simplified scope).

Theory Reference: Module 4

Program 8:
Title: Create a detailed dashboard that visually represents the energy consumption patterns.

Problem Description: Analyzing Energy Consumption Patterns in a Smart Building. Consider a dataset containing hourly energy consumption (in kilowatt-hours, kWh) for various building zones (e.g., offices, labs, common areas) over a year. The dataset also includes environmental data (temperature, humidity), occupancy data (estimated number of people in each zone), and time-related information (day of the week, month).

Method: Consider a Tableau data set, perform data preparation & exploration, visual analysis, advanced analysis & evaluation and data storytelling.

Theory Reference: Module 5

Suggested Learning Resources:

Textbooks:

1.	Elmasri and Navathe, Fundamentals of Database Systems, 7th Edition, Pearson Education, 2013.
2.	Cole Nussbaumer Knaflic, Storytelling with Data: A Data Visualization Guide for Business Professionals, John Wiley & Sons, Inc., Hoboken, New Jersey, 2015.

Reference Books:

1.	Abraham Silberschatz, Henry F. Korth, S. Sudarshan, Database System Concepts, 6th Edition, McGraw Hill,2010.
2.	Raghu Ramakrishnan and Johannes Gehrke, Database Management Systems, 3rd Edition, McGraw-Hill, 2013.

Web links and Video Lectures (e-Resources):
1. https://online.stanford.edu/courses/soe-ydatabases0001-databases-advanced-topics-sql

Course outcomes:

CO1	Use the appropriate high-performance database like parallel and distributed database.
CO2	Demonstrate the real-world data using object-oriented database.
CO3	Analyze the rule set in the database to implement data warehousing of mining.
CO4	Design the database for recent applications database for better interoperability.

CO-PO Mapping														
CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	-	-	-	1	1	-	3	3	2
CO2	3	2	3	2	2	-	-	-	1	1	-	3	3	2
CO3	3	2	3	2	2	-	-	-	1	1	-	3	3	2
CO4	3	2	3	2	2	-	-	-	1	1	-	3	3	2

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Project Phase-2	**Semester**	VII
Course Code	CS404	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L:T:P)	0:0:10	**Total**	100
Credits	5	**Exam. Duration**	---
Teaching Dept	CS		

The Final Year Project – Phase 2 focuses on the execution and completion of the project work as proposed and approved in Phase 1, under the continued guidance of the project guide. In this phase, students are expected to implement the planned solution or system, carry out necessary testing, analyze results, and refine their work based on feedback. Detailed documentation covering system design, implementation, testing procedures, outcomes, and conclusions must be prepared. The project culminates in a final presentation and demonstration before an evaluation panel. Successful completion of Phase 2 reflects the student's ability to apply theoretical knowledge to practical problems, demonstrating skills in research, development, and communication.

Course Title	Cyber Security	**Semester**	VII
Course Code	CS421	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L: T: P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview:
The Cyber Security (CS432) course covers fundamental cybersecurity principles, threats, and mitigation strategies. It includes cyber vigilance, reconnaissance techniques, spoofing, session hijacking, denial-of-service attacks, and buffer overflow vulnerabilities. Students will explore cybersecurity laws, ethical aspects, and real-world case studies. Practical activities involve analyzing attacks, simulating vulnerabilities, and applying security measures. The course equips students with the skills to protect digital systems and comply with cyber laws.

The course objective is to:

1. Understand cybersecurity fundamentals, cyber vigilance, and threats like spoofing, session hijacking, DoS, and buffer overflow.
2. Analyze reconnaissance techniques and their role in identifying vulnerabilities.
3. Apply cybersecurity practices and tools to mitigate attacks and improve vigilance.
4. Understand the legal and ethical aspects of cybersecurity, focusing on Indian cyber laws.

Prerequisite Knowledge: This course requires that the students are familiar with concepts of Computer Networks, and Cryptography.

Module-1: Cyber Security Fundamentals and Cyber Vigilance	**8 Hours**

Cyber Security Fundamentals: What is Cybercrime, Genesis and Evolution of Cybercrime and Cybersecurity, State of Cybersecurity Today, State of Cybersecurity in Organizations, Why Cybersecurity is not working as expected, Cybercrime and Vulnerability overview, Understanding the complexity of Cyber Protection.

Cyber Vigilance: Introduction to Cyber Vigilance: Cybersecurity & Cyber Vigilance compared, Understanding Cyber Vigilance, Mechanics of Data Sharing, What is Cyber Vigilance, Cyber Vigilance Mindset, What is cyber Safety, Cyber Vigilance Implementation and Organization.

Activity: Spot the Vulnerability
Analyze simple scenarios such as weak passwords or clicking suspicious links to identify vulnerabilities. Propose basic solutions to enhance cybersecurity, such as using strong passwords or avoiding phishing links. Discuss the findings with the class to understand how small actions can prevent cybercrime.

Textbook 1: 1, 2, 3, 4, 6, 8 and 9

Module-2: Reconnaissance	**8 Hours**

Reconnaissance: Introduction to Reconnaissance – Legal, Questionable, Illegal, and Impact of Reconnaissance, Social Engineering – Techniques, Physical Intrusion, Communication Media &

Countering Social Engineering, Internet Foot Printing – Social Networking, Web Searching, Network Enumeration, DNS – based & Network based Reconnaissance**.**

Activity:Network Reconnaissance Simulation

Use tools like ping or traceroute to map the path data takes through the network to a specific website or server. Identify the intermediate devices (routers, switches) in the path and discuss how attackers could exploit such reconnaissance techniques.

Textbook 2: Chapter 2.

Module-3: Spoofing & Session Hijacking	**8 Hours**

Spoofing: Process of an IP Spoofing, Costs of Spoofing, Types of Spoofing, Spoofing Tools, Prevention and Mitigation.

Session Hijacking: TCP Session Hijacking, Session Hijacking Tools, UDP Hijacking, Prevention and Mitigation.

Activity: Spoofing Demonstration and Discussion

Simulate an email spoofing scenario using a safe email header analysis tool (e.g., Gmail's "Show Original" feature). Analyze the headers of a spoofed email to identify red flags like mismatched sender addresses or unusual server paths. Discuss how spoofing can be used in phishing attacks.

Textbook 2: Chapters 7 & 8

Module-4: Denial Of Service Attacks & Buffer Overflow	**8 Hours**

Denial of Service Attacks: Causes for DoS Attacks, Types of DoS Attacks, Known DoS Attacks, Known DDoS Attacks, Prevention and Mitigation of DoS, and DDoS Attacks.

Buffer Overflow: Introduction, Types of Buffer Overflow, Detection and Prevention of Buffer Overflows.

Activity: DoS/DDoS Simulation and Mitigation

Simulate a simple DoS attack using a safe environment (e.g., sending multiple ping requests to a local server or test machine). Observe how the server slows down or becomes unresponsive. Discuss how this can escalate to a DDoS attack when multiple devices are used.

Textbook 1: Chapters 11 & 12

Module-5: Internet And Cyber Law	**8 Hours**

Internet and Cyber Law: Need for Cyber Law, India and Cyber Law, Internet Censorship, Cyber Crimes and Enforcement Agencies, IT Act: Aim and Objectives, Scope of the Act, Major Concepts, Digital Signature – Authentication of Electronic Records, Electronic Governance – Legal Recognition of Electronic Records and Digital Signatures, Use of Electronic Records and Digital Signatures in Government and Its Agencies, Retention of Electronic Records, Secure Electronic Records and Secure Digital Signatures.

Case Study: The 2016 Mirai Botnet Attack

Analyze the Mirai botnet attack, which exploited vulnerabilities in IoT devices to execute one of the largest DDoS attacks. Discuss how weak passwords, lack of cyber vigilance, and poor device security contributed to the attack. Explore reconnaissance techniques used to identify vulnerable devices, the role of IP spoofing in masking the source of the attack, and mitigation strategies like traffic filtering and proactive monitoring. Present recommendations to enhance cybersecurity and prevent similar attacks in the future.

Textbook 2: Chapters 19 & 20.

Advanced Learners:	
1	Investigate real-world examples of privilege escalation attacks, including token kidnapping. https://thecybersecguru.com/glossary/mastering-privilege-escalation-essential-guide/
2	Study the impact of phishing, smishing, and vishing on organizational security. https://scasecurity.com/blog/understanding-the-risks-and-mitigation-of-phishing-attacks/
3	Research advanced techniques in Intrusion Detection Systems (IDS) using Snort or Suricata. https://www.netgate.com/blog/suricata-vs-snort

Suggested Learning Resources:	
Textbooks:	
T1	Sudhir Goel, "Cyber Vigilance – The art of staying cyber safe", First Edition, Gurucool Publishing ,2023.
T2	Alfred Basta, Nadine Baste, Mary Brown, Ravinder Kumar, "Cyber Security and Cyber Laws", First Edition, Cengage, 2018.
Reference Books:	
R1	Nina Godbole, Sunit Belapure, "Cyber Security: Understanding Cyber crimes, Computer Forensics and Legal Perspectives", First Edition, Wiley India, 2011.
R2	Charles Pfleeger, Shari Pfleeger, Jonathan Margulies, "Security in Computing", Fifth Edition, Prentice Hall, New Delhi, 2015.
R3	Anand Shinde, "Introduction to Cyber Security Guide to the World of Cyber Security", 1st Edition, Notion Press, 2021

Course outcomes: Upon successful completion of this course, students will be able to	
CO1	Assess cyber threats, vulnerabilities, and principles of cyber vigilance.
CO2	Analyze reconnaissance techniques and their impact on cybersecurity.
CO3	Demonstrate basic tools and strategies to prevent and mitigate cyberattacks.
CO4	Apply cyber laws and ethical principles to enhance security and compliance.

CO/PO	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2						1				3		3
CO2	3	3		2								3		3
CO3	3	2	2		3							3	2	3
CO4	2	2				2		3				2		2

High-3 Medium-2 Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Distributed Computing & Block Chain	**Semester**	VII
Course Code	CS422	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview

This course introduces students to distributed systems, remote invocation, and consensus mechanisms. It covers distributed transactions, concurrency control, blockchain fundamentals, cryptography, Bitcoin mechanics, and decentralization. The course also explores Ethereum, smart contracts, Hyperledger, and IOTA, providing hands-on activities and case studies to reinforce learning.

Course objective is to:

- Understand the characterization of distributed systems, remote invocation
- Learn Coordination and Agreement, Consensus in group communication, distributed transaction and concurrency control.
- Explain the fundamentals of blockchain and Cryptography.
- To explore the driving force behind the cryptocurrency Bitcoin.
- Appreciate and demonstrate the Ethereum platform to develop blockchain application and familiarize with Smart contracts , Hyperledger and IOTA.

Prerequisite Knowledge: A Basic knowledge in Data structures, Cryptography, DBMS & Networking

Module-1	**08 Hours**

Characterization Of Distributed Systems: Introduction, Examples of distributed systems, **Remote Invocation**: Introduction, Request-reply protocols.
Coordination And Agreement: Introduction, Coordination and agreement in group communication, Consensus and related problems.

Activity: Case Study Discussion, Debate, MCQs

Textbook 1: Chapter 1.1,1.2, 5.1,5.2, 15.1, 15.4 &15.5

Module-2	**08 Hours**

Distributed Transactions: Introduction, Flat and nested distributed transactions, Atomic commit protocols, Concurrency control in distributed transactions.

Activity: Interactive Quiz, Concurrency Control Exercise: Conflict Detection

Textbook 1: Chapter-17.1-17.4

Module-3	**08 Hours**

Blockchain Basics: Introduction to blockchain, Types of blockchain, CAP theorem and blockchain, Benefits and limitations of blockchain. **Introduction to Cryptography & Cryptocurrencies:** Cryptographic Hash Functions, Hash Pointers and Data Structures, Digital Signatures, Public Keys as Identities, A Simple Cryptocurrency. **Activity**: Hash Function Challenge-Students manually calculate simple hash values for provided inputs, Hands-on Activity. **Textbook 3: Chapter 1.2-1.5, Textbook 2: Chapter 1**	
Module-4	**08 Hours**
How Bitcoin Achieves Decentralization: Consensus without identity using a blockchain, Incentives and proof of work **Mechanics of Bitcoin:** Bitcoin transactions, Bitcoin Scripts, Applications of Bitcoin scripts, Bitcoin blocks, The Bitcoin network, Limitations and improvements. **How to Store and Use Bitcoins:** Simple Local Storage, Hot and Cold Storage, Splitting and Sharing Keys, Online Wallets and Exchanges, Payment Services, Transaction Fees.. **Activity**: Simulate the Paxos consensus algorithm, Hot vs. Cold Storage Debate, Bitcoin Transaction Flow Diagram. **Textbook 2: Chapter 2.2-2.3, 3, 4**	
Module-5	**08 Hours**
Smart Contracts and Ethereum : Smart Contracts: Definition, Ricardian contracts. **Ethereum:** Introduction, Ethereum blockchain, Elements of the Ethereum blockchain, Precompiled contracts, Hyperledger Fabric model, IOTA. **Activity**: Ethereum Blockchain Visual Map:Provide students with blank diagrams of the Ethereum blockchain structure & make them fill, Quiz **Textbook 3: Chapter 10, M1-Week-5-Lecture 20, Week-7-Lecture-25**	
Follow on courses: DeFi and Smart Contracts, Advanced Blockchain Development, Ethereum and Solidity.	
Suggested Learning Resources: **Textbooks:**	
1.	George Coulouris, Jean Dollimore and Tim Kindberg, Distributed Systems Concepts and Design, Fifth Edition, Pearson Education, 2012.
2.	Arvind Narayanan, Joseph Bonneau, Edward W. Felten, Andrew Miller, Steven Goldfeder and Jeremy Clark., Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction. Princeton University Press, 2016.
3.	Imran Bashir, Packt Publishing Ltd, Mastering Blockchain - Distributed ledgers, decentralization and smart contracts explained, Second Edition.
Reference Books:	
1	Andreas M. Antonopoulos, Mastering Bitcoin: Unlocking Digital Cryptocurrencies, First Edition, O'Reilly Media, 2014
2.	Daniel Drescher, Blockchain Basics: A Non-Technical Introduction in 25 Steps, First Edition, Apress, 2017

Web links and Video Lectures (e-Resources):
M1: https://onlinecourses.nptel.ac.in/noc20_cs01/preview **M2:** https://nptel.ac.in/courses/106/105/106105184/
Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning Quiz, Group Discussions, Seminars, In class assignments.

Course outcomes:	
CO1	Analyze the performance of mutual exclusion, consensus algorithms.
CO2	Outline the fundamental concepts and algorithms related to distributed transactions.
CO3	Prepare the types, benefits and limitation of blockchain and explore the blockchain decentralization and cryptography concepts.
CO4	Demonstrate the principles and methodologies used in Bitcoin.
CO5	Design smart contracts, explore Ethereum, and understand Hyperledger and IOTA.

CO-PO Mapping

CO /PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	2	-	-	3	1	-	-	-	-	1	-	1
CO2	3	2	2	-	-	1	2	-	-	-	-	2	-	3
CO3	3	2	2	-	-	3	1	-	-	-	-	2	-	2
CO4	3	2	2	-	-	2	2	-	-	-	-	1	-	3
CO5	3	2	-	-	-	-	-	-	-	-	-	1	-	3

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Deep Learning	**Semester**	VII
Course Code	CS423	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course overview:

This Deep Learning course provides a comprehensive exploration of neural network architectures, training methodologies, and practical applications. It begins with foundational concepts, establishing a firm understanding of neural network basics, including neurons, activation functions, and feedforward networks. The course then delves into the intricacies of training deep neural networks, covering essential optimization techniques to prevent overfitting. The curriculum culminates in applying deep learning to real-world problems in computer vision, natural language processing, and speech recognition, while also touching upon advanced topics like reinforcement learning.

Course objective is to:

1. Introduce the fundamental concepts of deep learning and neural network architectures.
2. Develop the ability to build and apply deep learning models for image-related tasks using CNNs and transfer learning.
3. Explore sequence modeling and text processing using RNNs, LSTMs, and Transformers.
4. Implement generative models and advanced training techniques.
5. Expose students to model deployment practices, interpretability, and ethical aspects of AI.

Prerequisites: Machine Learning

Module-1: Fundamentals of Deep Learning and Neural Networks	**Hours: 8**

Introduction to deep learning, Applications: Image classification, object detection, image segmentation, Text generation, translation, sentiment analysis, Automatic speech recognition (ASR), text-to-speech (TTS), Tensors and tensor operations, Basics of neural networks, Forward and backward pass, Training, loss functions, and optimizers

Activity: Build and train a basic classifier on MNIST/Fashion-MNIST

Textbook 1: Chapter 1-3

Module-2: Computer Vision and Convolutional Neural Networks	**Hours: 8**

Convolutional layers, pooling, and deep CNNs, Data augmentation and regularization, Transfer learning and fine-tuning with pretrained models (e.g., VGG, ResNet).

Activity: 1. Train a CNN on a custom image dataset
2. Fine-tune a pretrained model on a new task

Textbook 1: Chapter 5-6

Module-3: Text and Sequence Modeling with RNNs and Transformers	**Hours: 8**

Word embeddings (Word2Vec, GloVe), Recurrent Neural Networks (RNNs), LSTM, GRU, Attention mechanism and Transformer introduction, Sequence-to-sequence models
Activity: 1. Build an LSTM-based sentiment analysis model 2. Create a basic Transformer for text classification **Textbook 1:** Chapter 9-11

Module-4: Advanced Deep Learning Techniques	**Hours: 8**
Model evaluation and hyperparameter tuning, Generative models: Autoencoders, Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), Custom training loops with train_step in Keras. **Activity**: 1. Develop a VAE for image reconstruction 2.Build a simple GAN on the MNIST dataset **Textbook 1:** Chapter 7,8,12	

Module-5: Deployment, Explainability, and Ethics	**Hours: 8**
Model deployment using TensorFlow SavedModel, TFLite, Interpretability: SHAP, LIME, saliency maps, Fairness and bias in deep learning, Responsible AI and ethical considerations. **Activity**: Analyze a model for bias and explainability **Textbook 1:** Chapter 13-14	

Suggested Learning Resources: **Textbooks:**	
1.	*Deep Learning with Python*, François Chollet, 2nd Edition, Manning Publications, 2021.
Reference Books:	
1.	"Neural Networks and Deep Learning" by Michael Nielsen, 1st edition, Determination Press 2015.
Web links and Video Lectures (e-Resources): • https://www.tensorflow.org/, https://pytorch.org/,https://keras.io/	
Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning The students with the help of the course teacher can take up relevant technical –activities like Research or presentation or demonstration of workflow/ implementation of an algorithm which will enhance their skills. The prepared report/research paper shall be evaluated for CIE marks.	
Course outcomes:	
CO1	Illustrate and implement basic neural networks, activation functions, and optimizers using TensorFlow/Keras.
CO2	Apply convolutional neural networks and transfer learning to solve real-world image classification tasks.
CO3	Design and train models for natural language processing using RNNs, LSTMs, and Transformer architectures.
CO4	Implement generative models (Autoencoders, GANs) and optimize training workflows.
CO5	Utilize the models, interpret model predictions, and evaluate models for fairness and ethical compliance.

CO PO Mapping														
CO/ PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PSO 1	PS O2	PS O3
CO1	3	2	1	1	3	0	0	1	0	1	0	3	2	0
CO2	3	3	2	2	3	0	0	1	1	1	0	3	2	0
CO3	3	3	3	3	3	1	0	1	2	2	1	3	2	0
CO4	3	2	3	2	3	1	1	1	2	3	2	3	2	0
CO5	3	3	3	3	3	2	1	1	2	3	3	3	3	0

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Computer Vision and Robotics	**Semester**	VII
Course Code	CS424	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview

Computer Vision and Robotics typically covers the fundamentals of both fields, including image processing, object recognition, machine learning for visual data, and robotics principles, culminating in hands-on projects integrating vision and robotics.

Course objective is to:

- To introduce students the fundamentals of image formation
- To introduce students the major ideas, methods, and techniques of computer vision and pattern recognition;
- To develop an appreciation for various issues in the design of computer vision and object recognition systems;
- To provide the student with programming experience from implementing computer vision and object recognition applications.

Prerequisite Knowledge: This course requires that the students are familiar with Image Processing

Module-1: Introduction	**Hours 8**

Image Formation-Geometric Primitives and transformation, photometric image formation, digital camera, point operators, more neighborhood operators, Interpolation, Decimation, Multi resolution representation, Geometric Representation.

Activity: Create a Basic Object Detector/Tracker

Textbook1: Chapter 2 2.1-2.3 , Chapter 3 3.1-3.6

Module-2: Object Recognition	**Hours 8**

Image Recognition, Image classification – Feature based methods. Object Detection- Face detection , Pedestrian detection, General object detection. Semantic segmentation, Application: Medical image segmentation, Instance segmentation, Panoptic segmentation, Application: Intelligent photo editing, Pose estimation.

Activity: Using Computer Vision perform a variety of facial applications, including facial recognition, building a virtual makeover system.

Textbook1: Chapter 6 6.1’6.2 ,6.3 and 6.4

Module-3: Feature Detection and matching	**Hours 8**

Points and patches - Feature detectors, Feature descriptors, Feature matching, Large-scale matching and retrieval, Feature tracking, Application: Performance-driven animation, Edges and contours, contour tracking- Level Sets, Lines and vanishing points- Successive approximation, Hough transforms, Vanishing points, segmentation- Graph-based segmentation.

Activity: Develop human activity using Computer Vision on the Raspberry Pi

Textbook1: Chapter 7 7.1,7.2, 7.3, 7.4 and 7.5	
Module-4: Motion Estimation	**Hours 8**
Translation alignment- Hierarchical motion estimation, Fourier-based alignment, Incremental refinement . Parametric motion- Application: Video stabilization,Spline-based motion. Optical Flow-Multi frame motion, Layered Motion – Frame Interpolation, Video Object Segmentation, Video Object tracking. Structure from Motion, Two frame structure from motion. Muti frame structure from motion. Simultaneous localization and mapping (SLAM). **Activity**: Build Image Search Engine using CBIR **Textbook1:** Chapter 9 9.1,9.2,9.3 and 9.4 **Textbook1:** Chapter 11 11.1, 11.2.11.3 and 11.4	
Module-5: Foundation of Robotics	**Hours 8**
Introduction – Components, Degrees of Freedom, Joints, Coordinates, Mechanisms, Controller. Kinematics, Dynamics, Position and Orientation of Objects, Coordinate Transformation. **Activity**: Design of underwater robot for surveillance purpose **Textbook 3:** Chapter 1 1.7, 1.8 Chapter 2 2.2, 2.3,2.4 2.5	

Suggested Learning Resources: **Textbooks:**	
T1	Richard Szeliski, Computer Vision: Algorithms and Applications, 2nd Edition, Springer, 2022.
T2	D. Forsyth and J. Ponce, Computer Vision-Amodern approach, Prentice Hall Robot Vision, by B .K.P.Horn, McGraw-Hill, 2011.
T3	Saeed B Niku, Introduction to Robotics Analysis, Control, Applications, 3rd Edition, Wiley, 2020.
Reference Books:	
R1	Haralick&Shapiro, Computer and Robot Vision, Vol II, 1st Edition, Addison-Wesley, 1990.
R2	Gerard Medioni and Sing Bing Kang, Emerging Topics in Computer Vision, 1st Edition, Prentice Hall PTR, 2004.
Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning Low cost Agri-Rover	

Course outcomes:	
CO1	Outline basic concepts, terminology, theories, models and methods in the field of computer vision
CO2	Identify known principles of human visual system
CO3	Identify basic methods of computer vision related to multi-scale representation, edge detection and detection of other primitives, stereo, motion and object recognition,
CO4	Design a computer vision system for a specific problem

CO-PO Mapping														
CO/PO	PO1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO2	PSO3
CO1	3	3	3	-	3	-	-	-	-	-	-	3	3	-
CO2	3	3	3	-	3	-	-	-	-	-	-	3	3	-
CO3	3	3	3	-	3	-	-	-	-	-	-	3	3	-
CO4	3	3	3	-	3	-	-	-	-	-	-	3	3	-

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Data Warehousing	Semester	VII
Course Code	CS425	Course Type	PEC
Total No. of Contact Hours	40	CIA Marks	50
No. of Contact Hours/week (L:T:P)	3:0:0	SEE Marks	50
Credits	3	Total Marks	100
Teaching Department	CS	Exam. Duration	3 hrs

Course Overview: This course introduces the fundamental concepts, methodologies, and technologies used in Data Warehousing. It covers the evolution of data warehousing, its architectural components, project management practices for data warehouse development, metadata management, and the role of OLAP, ETL processes, and data quality. Students will gain hands-on experience in designing, implementing, and managing data warehouses, as well as in deploying tools for information delivery and OLAP.

Course objectives : The student will be able to:

- Understand the need of data warehousing.
- Understand the planning a data warehouse based on business requirements
- Understand the architectural components of Data warehouse
- Understand the data modeling approaches in Data Warehousing
- Understand OLAP operations and use them effectively to improve data quality

Prerequisites: Basic Knowledge of Databases, Foundations in Data Structures and Algorithms

Module-1	**8 Hours**

Escalating Need for Strategic Information, Failures Of Past Decision-Support Systems, Operational Versus Decision-Support Systems, Data warehousing—The Only Viable Solution, Data Warehouse Defined. The Data warehousing Movement, Evolution of Business Intelligence Data Warehouse: The Building Blocks: Defining Features, Data Warehouses and Data Marts, Architectural Types, Components: Source Data Component, Data Staging Component, Data Storage Component, Information Delivery Component, Metadata Component, Management and Control Component, Metadata In The Data Warehouse.

Self-Learning Activity: Research on the evolution of business intelligence tools and their impact on decision-making. Prepare a timeline or a comparative study of the past vs. modern BI systems.

Text Book: Chapter 1 , 2

Module-2	**8 Hours**

Planning And Project Management: Planning Your Data Warehouse, The Data Warehouse Project, The Development Phases, The Project Team, Project Management Considerations Defining The Business Requirements: Dimensional Analysis, Information Packages: Requirements Not Fully Determinate, Business Dimensions, Dimension Hierarchies and Categories, Key Business Metrics Or Facts, Requirements Gathering Methods, Data Sources, Data Transformation, Data Storage, Information Delivery, Information Package Diagrams. Requirements As the Driving Force For Data warehousing Data Design, The Architectural Plan, Data Storage Specifications, Information Delivery Strategy.

Self-Learning Activity: Create a sample project plan for implementing a data warehouse, outlining the stages, timelines, team structure, and resources required. **Text Book: Chapter 4, 5,6**	
Module-3	**8 Hours**
Architectural Components: Understanding Data Warehouse Architecture, Distinguishing Characteristics, Architectural Framework, Technical Architecture, Architectural Types. Infrastructure As the Foundation for Data warehousing: Infrastructure Supporting Architecture, Hardware and Operating Systems, Database Software, Collection of Tools, Data Warehouse Appliances. The Significant Role of Metadata: Why Metadata Is Important, Metadata Types by Functional Areas, Business Metadata, Technical Metadata, How to Provide Metadata. **Self-Learning Activity:** Investigate and compare different data warehouse architectures (e.g., 3-tier vs. 2-tier). Document their pros, cons, and use cases in a tabular format. **Textbook: Chapter 7,8,9**	
Module-4	**8 Hours**
Principles Of Dimensional Modelling: From Requirements to Data Design, The Star Schema, Star Schema Keys, Advantages of the Star Schema, Star Schema: Examples, Dimensional Modelling: Advanced Topics: Updates to The Dimension Tables, Miscellaneous Dimensions, The Snowflake Schema, Aggregate Fact Tables, Families of Stars. Data Extraction, Transformation, And Loading: ETL Overview, ETL Requirements and Steps, Data Extraction, Data Transformation, Data Loading, ETL Tool Options Reemphasizing ETL Metadata, ETL Summary and Approach. **Self-Learning Activity:** Design a star schema for a retail business data warehouse, defining facts, dimensions, and relationships. Use an ETL tool (e.g., Talend) to map a simple ETL process. **Text Book: Chapter 10,11,12**	
Module-5	**8 Hours**
Data Quality: A Key to Success: Why Is Data Quality Critical? Data Quality Challenges, Data Quality Tools, Data Quality Initiative, Master Data Management (Mdm). Matching Information to The Classes of Users: Information from the Data Warehouse, Who Will Use the Information? Information Delivery. Information Delivery: Business Activity Monitoring (Bam), Dashboards and Scorecards OLAP In the Data Warehouse: Demand for Online Analytical Processing, Major Features and Functions, OLAP Models, OLAP Implementation Considerations. Data Warehousing and the Web: Web-Enabled Data Warehouse, Web-Based Information Delivery, OLAP And the Web, Building A Web-Enabled Data Warehouse. **Self-Learning Activity:** Analyze a set of data from an open-source BI platform (e.g., Tableau public) and perform data quality checks on it. Create a basic dashboard for reporting. **Textbook: Chapter 13,14,15,16**	
Open ended experiments leading to guided projects (Beyond syllabus): • Design a Mini Data Warehouse for E-Commerce • Build an ETL Pipeline using Python or Talend • Develop OLAP Cubes and Dashboards • Implement Metadata Repository using MySQL/PostgreSQL	
Suggested Learning Resources: Textbooks:	
1	Paulraj Ponniah, Data Warehousing Fundamentals for IT Professionals, Second Edition, Wiley 2010.
Reference Books:	
1	S. K. Singh, Data Warehousing: Concepts, Techniques, Products and Applications, Pearson Education, 1st Edition,2006.

Web links and Video Lectures (e-Resources):
• https://onlinecourses.swayam2.ac.in/cec19_cs01/preview • http://nptel.ac.in/video.php?subjectId=106106093
Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning ● Course Project - Build a simple data warehouse using various schemas**.**

Course outcomes: At the end of the course, the student will be able to	
CO1	Identify and outline data warehousing evolution and architectural components.
CO2	Plan, design, and prepare data warehouse architecture and requirements.
CO3	Analyze and evaluate technical architecture and metadata in data warehouses.
CO4	Create dimensional models and demonstrate ETL processes effectively.
CO5	Assess data quality, critique MDM approaches, and evaluate OLAP solutions.

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2			2							2	2	
CO2	2	2	3	3	2							3	2	
CO3		3		3	3							3		3
CO4			3		2					2		3	3	
CO5		3			2			2		2		2	3	

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Fuzzy Logic	**Semester**	VII
Course Code	CS426	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview:
This course presents the methods to represent uncertainty within engineering problems. The course helps to understand fundamental concepts of fuzzy logic and fuzzy set theory. It provides techniques to design and implement fuzzy logic systems. The course also provides techniques for classification in the presence of fuzziness.

Course objective is to:

1. Learn about methods to quantify uncertainty within engineering.
2. Understand the difference between classical sets and fuzzy sets
3. Learn the fundamental concepts of fuzzy logic and apply them to different domains
4. Learn the techniques used in development of fuzzy systems
5. Understand fuzzy techniques for applications like classification and clustering

Prerequisite Knowledge: Understanding of set theory and relational algebra.

Module-1	**Hours: 8**

Introduction, Classical Sets and Fuzzy Sets: Introduction, Classical Sets - Operations on Classical Sets, Properties of Classical (Crisp) Sets, Mapping of Classical Sets to Functions: **Fuzzy Sets -** Fuzzy Set Operations, Properties of Fuzzy Sets, Alternative Fuzzy Set Operations.

Activity: Take a few real-world examples and develop fuzzy sets for them.

Textbook 1: Ch. 1,2

Module-2	**Hours: 8**

Classical Relations and Fuzzy Relations: Cartesian Product, Crisp Relations, Fuzzy Relations, Tolerance and Equivalence Relations, Fuzzy Tolerance and Equivalence Relations, Value Assignments

Activity: Take the example of a product such as video multiplexer and develop the fuzzy relations.

Textbook 1: Ch. 3

Module-3	**Hours: 8**

Properties of Membership Functions, Fuzzification and Defuzzification: Features of the Membership Function, Various Forms, Fuzzification, Defuzzification to Crisp Sets, Lambda - Cuts for Fuzzy Relations, Defuzzification to Scalars.
Logic: Classical Logic, Proof, Fuzzy Logic, Approximate Reasoning, Other Forms of the Implication Operation

Activity: Do fuzzification and defuzzification of a solid waste treatment and disposal system.

Textbook 1: Ch. 4, 5 (Part 1)

Module-4	**Hours: 8**
Fuzzy Systems: Natural Language, Linguistic Hedges, Fuzzy Rule Based Systems, Graphical Techniques of Inference. **Development of Membership Functions**: Membership Value Assignments (Intuition, Inference, Rank Ordering, Neural Networks, Genetic Algorithms) **Activity**: Develop fuzzy membership functions for a controller to regulate the temperature of a room. **Textbook 1: Ch. 5 (Part 2), 6**	
Module-5	**Hours: 8**
Fuzzy Classification: Classification by Equivalence, Relations - Crisp Relations, Fuzzy Relations, Cluster Analysis, Cluster Validity, cMeans Clustering - Hard c-Means (HCM), Fuzzy c-Means (FCM), Classification Metric, Hardening the Fuzzy c-Partition, Similarity Relations from Clustering. **Activity**: Take a real world problem and apply fuzzy c-means clustering. Implement fuzzy clustering using python package *skfuzzy*. **Textbook 1: Ch. 10**	

Suggested Learning Resources: **Textbooks:**	
1	Timothy J. Ross, Fuzzy Logic with Engineering Applications, 3rd Edition, Wiley India ,2010.
Reference Books:	
1	George J. Klir, Bo Yuan, Fuzzy Sets and Fuzzy Logic: Theory and Applications, 1st edition, Prentice Hall, 1995.
2.	J.S.R. Jang, C.T. Sun and E. Mizutani, Neuro-Fuzzy and Soft Computing: A Computational Approach to Learning and Machine Intelligence, 1st edition, Pearson Education, 2015.

Web links and Video Lectures (e-Resources):

- **VTU online course:** Fuzzy Sets, Logic and Systems & Applications

https://online.vtu.ac.in/course-details/Fuzzy-Sets-Logic-and-Systems-Applications

- **NPTEL Course**: Fuzzy Sets, Logic and Systems & Applications
  https://onlinecourses.nptel.ac.in/noc20_ee03/preview

https://www.youtube.com/watch?v=K7S3TgfqnX0

- **NPTEL Course**: Fuzzy Logic and Neural Networks.
  https://onlinecourses.nptel.ac.in/noc22_ge04/preview

Course outcomes:	
CO1	Utilize fuzzy sets and fuzzy logic operations.
CO2	Outline the difference between classical operations and fuzzy operations
CO3	Apply the methods for fuzzification and defuzzification
CO4	Design logic systems and fuzzy systems

CO5	Apply fuzzy techniques for classification and clustering

CO-PO Mapping

CO/ PO	PO1	PO2	PO3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	1	-	-	-	-	-	-	2	-	-
CO2	3	2	2	1	1	-	-	-	-	-	-	2	-	-
CO3	3	2	2	1	1	-	-	-	-	-	-	2	-	-
CO4	3	2	2	2	1	-	-	-	-	-	-	2	-	-
CO5	3	2	2	2	2	-	-	-	-	-	-	2	-	-

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	DevOps	**Semester**	VII
Course Code	CS427	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week(L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview: DevOps course is designed to introduce learners to essential tools, methodologies, and practices that form the backbone of modern software development and deployment pipelines. The course covers foundational DevOps principles and progresses through hands-on exposure to industry-standard tools including Git, Docker, Kubernetes, Jenkins, Ansible, Prometheus, Grafana, and Terraform.

Course objective is to:

1. Understand the Core Concepts of DevOps.
2. Apply CI/CD pipelines using industry-standard tools.
3. Design and develop containerized and orchestrated applications with automated infrastructure
4. Manage Infrastructure and Configuration.

Prerequisite Knowledge: This course requires that the students are familiar with basic Programming Concepts, Scripting Languages, Operating Systems and Networking.

Module-1: Introduction to DevOps, Git	**8 Hours**

Introduction to Devops, SDLC, History and evolution of DevOps, Key concepts and principles, DevOps lifecycle, Importance and benefits of DevOps. Version control system and its types, Introduction to GIT, GIT Basics commands (Creating Repositories, clone, push, commit, review), Git workflows- Feature workflow, Master workflow, Centralized workflow, Feature branching, Managing Conflicts, Tagging and Merging, Best Practices- clean code.

Activity: Hands-on exercises with Git, Creating and managing a repository.

Textbook 1: Chapter 1,2.

Module-2: Docker and Kubernetes	**8 Hours**

Introduction to containers, Docker architecture and components, Basic Docker commands (build, run, images), Dockerfile and containerization of applications, Docker Container Management.

Introduction to container orchestration, Kubernetes architecture, Basic Kubernetes objects (pods, services, deployments), Deploying applications on Kubernetes, Kubernetes Application Deployment.

Activity: Installing Docker, Creating and running Docker containers, Setting up a Kubernetes cluster, Deploying a sample application.

Textbook 2: Chapter 1,2,3,4,5,6,7

Module-3: Jenkins and Ansible	**8 Hours**

Introduction to CI/CD, Setting up Jenkins, Creating and running basic Jenkins jobs, Integrating Jenkins with GitHub, CI Pipeline Configuration. Introduction to configuration management, Basics of Ansible, Writing and running Ansible playbooks, Managing infrastructure with Ansible, Ansible Playbook Creation. **Activity:** Jenkins installation and setup, Configuring a simple CI pipeline. Installing and configuring Ansible, Writing simple playbooks. **Textbook 1:** Chapter 3.

Module-4: Prometheus, Grafana	**8 Hours**
Importance of monitoring and logging in DevOps, Introduction to Prometheus and Grafana, Setting up monitoring with Prometheus, Visualizing metrics with Grafana, Prometheus and Grafana Configuration **Activity**: Installing Prometheus and Grafana, Configuring basic monitoring and alerts. **Textbook 3: Chapter 7**	

Module-5: Terraform	**8 Hours**
Introduction to IaC, Basics of Terraform, Writing and applying Terraform configurations, Managing infrastructure with Terraform, Terraform Scripting. **Activity**: Installing Terraform, Creating and applying Terraform scripts **Textbook 1:** Chapter 11	

Follow on Courses: Advanced DevOps Specializations like DevSecOps practices , Containerization and Orchestration.
Resources for Advanced Learners: https://courses.networknuts.net/kubernetes-lp/?gad_source=1&gclid=CjwKCAiAm-67BhBlEiwAEVftNg_rF2nR_aiO4thwxWGB3ryF4g0uJmr-2zL9mSbNdNqB90VAwspSdhoCGgUQAvD_B

Suggested Learning Resources: **Textbooks:**	
1.	Gene Kim, Patrick Debois, John Willis, and Jez Humble, The DevOps Handbook: How to Create World-Class Agility, Reliability, & Security in Technology Organizations, 1st Edition, IT Revolution Press, 2016.
2.	Emily Freeman, DevOps for Dummies,1st Edition, John Wiley & Sons, 2019
Reference Books:	
1.	Len Bass, Ingo Weber, and Liming Zhu, DevOps: A Software Architect's Perspective,1st Edition, Addison- Wesley Professional, 2015.
2.	Gene Kim, Patrick Debois, John Willis, Jez Humble, John Allspaw, The DevOPS Handbook: How to Create World-Class Agility, Reliability, and Security in technology Organizations, IT Revolution Press, 2016.

Course outcomes:	
CO1	Identify the need for DevOps and list down the primary benefits of DevOps from perspective of business and IT teams.
CO2	Outline the ways in which DevOps uses new tools/technologies to deliver quality software more rapidly.
CO3	Demonstrate the practices of version control and configuration management.
CO4	Identify the essentials of continuous integration (CI) and outline the principles and practices of continuous delivery (CD) .
CO5	Implement an automated deployment pipeline and create a DevOps toolchain.

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	-	-	-	1	1	-	3	3	2
CO2	3	2	3	2	2	-	-	-	1	1	-	3	3	2
CO3	3	2	3	2	2	-	-	-	1	1	-	3	3	2
CO4	3	2	3	2	2	-	-	-	1	1	-	3	3	2
CO5	3	2	3	2	2	-	-	-	1	1	-	3	3	2

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Software Product Management	**Semester**	VII
Course Code	CS428	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview:
This course introduces the principles and practices of Software Product Management, focusing on the end-to- end management of software products. Students will learn to align product development with business goals, collaborate across teams, and employ strategic management techniques to drive innovation, ensure customer satisfaction, and deliver successful products to the market. Through real-world examples and practical
applications, the course prepares students to effectively oversee the product lifecycle and make data-driven decisions in a competitive software industry.

Course objective is to:

1. Understand the nuances of software products business
2. Understand the pivotal role of Software Product Manager
3. Understand the orchestration between product management and other business functions
4. Understand the global best practices in Software Product Management

Prerequisite Knowledge: This course requires that students are familiar with software product management concepts, software development processes, and have experience working with software-intensive systems.

Module-1 Introduction and Foundations	**8 Hours**

Introduction and Foundations, and Software Product Management (SPM) Essentials: Introduction to the course, Software as a Business, SPM Framework, The Role and organization of SPM.
Software Product Management Framework: Overview, Product Management Definitions, Product Family, Product Line, Product Platform, Structure of the SPM Framework and Startup SPM Framework.

Activity: Understand the concepts of product family, product line, and product platform.
Provide a scenario as: "Your team is creating a line of educational apps for students across different age groups." Ask students to list possible products in the family, common features (platform), and differentiating features for each product and present their ideas briefly.

Text Book : Ch2[2.1,2.2,2.3,2.5,2.6]

Module-2 Business Models, Product Strategy	**8 Hours**

Business Models, Product Vision, Product Name, Product Definition, Positioning, Service Strategy, Sourcing, Pricing, Financial Management, Ecosystem Management, Legal Aspects, Performance and Risk Management, and Product Strategy Processes and Documentation

Activity: Visualize a product's ecosystem.

Provide an example product (e.g., a gaming console). Ask students to create a map showing key ecosystem players (e.g., developers, distributors, and users) and their interactions. Discuss how managing these relationships impacts success.

Text Book : Ch.2,3 [2.4, 3.1,3.2,3.5,3.7-3.15]

Module-3 Product Planning	**8 Hours**
Customer Insight, Product Planning Approaches, Release Management, Roadmapping, Product Life Cycle Management and Process Measurement and Improvement. **Activity:** Understand customer insights. Provide a fictional product idea (e.g., a fitness tracking app). Ask students to create a customer persona, including: Demographics (age, profession, etc.), Goals and pain points, How the product addresses their needs and Groups briefly present their personas and discuss how these insights influence product planning. **Text Book: Ch. 4**	
Module-4 Strategic Management	**8 Hours**
Corporate Strategy, Portfolio Management, Innovation Management, Resource Management, Compliance Management, Market Analysis, Product Analysis, Corporate Strategy Processes. **Activity:** Innovation Pipeline Creation: Understand innovation management. Provide a scenario: A company in need of innovation to stay competitive (e.g., an electric vehicle manufacturer). Ask students to brainstorm 2-3 innovative product ideas and outline a simple pipeline for development (concept, prototype, testing, launch). Groups present their pipelines. **Text Book: Ch. 5**	
Module-5 Orchestration of the Organization's Functional Areas	**8 Hours**
Role and Process, Development and UX Design, Marketing, Sales and Fulfilment, Delivery Service and Support, and Orchestration Skills. **Activity:** Marketing Message Creation: Craft an effective marketing message. Assign a fictional product (e.g., an eco-friendly task management app). Ask students to create a tagline and a short promotional message that highlights its unique value. Groups share their messages, and the class discusses which are most compelling. **Text Book: Ch.6**	

Follow-on courses: User Experience Design and Strategy, Digital Marketing and Analytics, Agile Product Management Practices.
Suggested Learning Resources: **Textbooks:**
Software Product Management for Start-Ups – The ISPMA®-Compliant Handbook and Study Guide, 2nd edition, Springer, 2022.
Reference Books:
Building Products for the Enterprise: Product Management in Enterprise Software" by Blair Reeves and Benjamin Gaines. 1st edition, O'Relly, 2018

Group Project - details

1. As part of the course requirements, students (in groups of up to four) will study a software product or company of their choice and apply the concepts learned in the Software Product Management course to gain insights into the company's products.
2. The project team will interact with the company's product leadership, ideally once a week, to connect class learnings with real-world product scenarios and draw conclusions. (Recommended)
3. Students shall submit a Report/slide presentation of findings & recommendations to the company – and present the final presentation in the class.
4. Evaluation Criteria and Weightage:
    1. 1/3rd for Comprehension of Concepts and their application
    2. 1/3rd for Completeness /Coverage of frameworks /methods learned in the course
    3. 1/3rd for Quality of Recommendations and articulation of best practices in product management.

Course outcomes:	
CO1	Identify key concepts and strategies for analyzing software product business dynamics in competitive markets.
CO2	Demonstrate the role of a Software Product Manager and their effective collaboration with key business functions, including marketing, sales, research, development, and support.
CO3	Analyze global best practices in Software Product Management to comprehend their influence on product lifecycle management.
CO4	Apply strategic management and portfolio techniques to drive innovation, ensure compliance, and optimize market and resource analysis.
CO5	Analyze the role of a Software Product Manager in orchestrating cross-functional business operations to drive product success and innovation.

CO-PO Mapping

CO/ P O	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 1 0	PO 1 1	PS O 1	PS O 2	PS O3
CO1	-	3	-	-	-	-	-	-	-	-	2	-	2	-
CO2	-		2	-	-	-	-	-	3	3	-	-	2	-
CO3	-	-	-	2	2	-	-	-	-	-	3	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	2	-
CO5	-	-	2	2	2	-	-	1	3	-	3		3	3

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Advances in Software Architecture	**Semester**	VII
Course Code	CS429	**CIA**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview: This course provides an in-depth exploration of modern software architecture principles, patterns, and technologies. It covers advanced topics such as micro-services, cloud-native architectures, DevOps, event-driven design, and AI-driven automation. Students will learn to design and evaluate scalable, secure, and maintainable software systems, applying industry best practices and emerging trends.

Course objective is to:

1. Understand the fundamentals of software architecture.
2. Understand the various software modeling techniques.
3. Understand software implementation and deployment
4. Learn the architecture of different applications.

Prerequisite Knowledge: This course requires that the students are familiar with programming (Java, Python, or C++), data structures, and algorithms.

Module-1 Fundamental Concepts and Architectural Design Principles	**8 Hours**

Understanding Software Architecture-Concepts of Software Architecture, Models, Processes, Stakeholders, the Design Process, Architectural Conception, Refined Experience in Action: Styles and Architectural Patterns, Architectural Conception in Absence of Experience.

Activities:

1. Identifying the pitfalls that are likely to occur for software architecture teams.
2. Discussing about the role of Software architects as Cost estimator**s.**

Text Book : Ch1, 1.1-1.13

Module-2 Connectors, Modeling	**8 Hours**

Connectors in Action: A Motivating Example, Connector Foundations, Connector Roles, Connector Types and Their Variation Dimensions, Example Connectors, Modeling Concepts, Ambiguity, Accuracy, and Precision, Complex Modeling: Mixed Content and Multiple Views, Evaluating Modeling Techniques, Specific Modeling Techniques.

Activity:

Identifying a few commercial products which are infrastructure components that provide asynchronous messaging service.

Finding out the different possibilities of splitting the system into a number of computationally independent execution structures

Text Book : Ch.3 3.1-3.8

Module-3 Analysis, Implementation And Deployment	**8 Hours**

Concepts, Existing Frameworks, Software Architecture and Deployment, Software Architecture and Mobility, Pipes and Filters, Event- based, Implicit Invocation, Layered systems, Repositories

Interpreters, Process control. **Activity:** Identifying the type of a given architectural pattern. Representing software using pipe-filter architecture. **Text Book : Ch.3, 3.9-3.17**

Module-4 Applied Architectures And Styles	**8 Hours**
Distributed and Networked Architectures, Architectures for Network-Based Applications, Decentralized Architectures, Service-Oriented Architectures and Web Services, Efficiency, Complexity, Scalability and Heterogeneity, Adaptability, Dependability. **Activity:** Identifying functional aspects of a service oriented architecture. Discussing the pros and cons of implementing a middle ware to deal with architectural issues. **Text Book : ch.4, 4.1-4.1**	

Module-5 Implementation	**8 Hours**
Understanding quality attributes- Availability- Deployability- Working with Other Quality Attributes- Virtualization- the Cloud and Distributed Computing- Architecturally Significant Requirements- Designing an Architecture**.** **Activity:** Identifying the cost of modifications in projects that measure deployment separately. Using the cloud assumes your application is service oriented. Find some examples that would support that statement and, if it is not universally true, find some that would falsify it. **Text Book: Ch.5, 5.1-5.9**	

Suggested Learning Resources: **Textbooks:**	
T1	Richard N. Taylor, Nenad Medvidovic, and Eric Dashofy, Software Architecture: Foundations, Theory, and Practice, Wiley Publications, 2009.
Reference Books:	
R1	Richard N. Taylor, Nenad Medvidovic, Eric Dashofy, Software Architecture: Foundations, Theory, and Practice, 2009.
R2	Steven John Metsker, Design Pattern Java Workbook, Addison Wesley Workbook, 2002.
Web links and Video Lectures (e-Resources): 1. https://onlinecourses.nptel.ac.in/noc19_cs65/preview	
Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning Students may be given problem domain that they may be encouraged to come out with multiple solution domains by applying some pattern. The best solution would be selected and presented.	

Course outcomes:	
CO1	Develop Software applications starting from software architecture and design.
CO2	Apply different types of systems analysis techniques and software design strategies.
CO3	Implement and deploy software applications.
CO4	Evaluate and implement different types of design patterns based on the requirement and

CO-PO Mapping															
CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	2	-	-	-	-	-	-	-	3	3	2
CO2	3	2	2	1	2	-	-	-	-	-	-	-	3	3	2
CO3	3	2	2	1	2	-	-	-	-	-	-	-	3	3	2
CO4	3	2	2	1	2	-	-	-	-	-	-	-	3	3	2

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Unix Shell Programming and Git	**Semester**	VII
Course Code	CS431	**CIE**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week L:T:P	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

Course Overview:
This course introduces Unix/Linux shell programming and Git-based version control. Students will learn to write shell scripts, manage processes, files, and users, and schedule jobs in a Linux environment. The course also covers essential Git operations like branching, merging, tagging, and history management. By the end, students will be able to automate system tasks and effectively use Git for individual and collaborative development.

Course objective is to:

1. Develop proficiency in using Unix/Linux shell commands for system and file management.
2. Understand and write shell scripts for task automation.
3. Manage users, processes, and jobs in a UNIX environment.
4. Introduce version control concepts using Git and demonstrate their practical use.
5. Apply Git for collaborative development, branching, merging, tagging, and history analysis.

Prerequisites: Basic understanding of Operating Systems. Familiarity with command-line interfaces. Prior exposure to programming languages such as C/PYTHON.

Module-1 Introduction to Shell Scripting	**Hours: 8**

Overview of Shell and Shell Scripting, Types of Shells, Creating and executing shell scripts, Input/output redirection, pipes, and filters, Basic commands: echo, read, print.

Activity: Automate Daily Greeting Script.

Textbook 1: 1,2,3,4,5,9,10,11

Module-2 System and File Management	**Hours: 8**

Process stages and types, Process management commands: ps, top, kill, nice, renice, pkill, jobs, fg, bg, Job scheduling: cron, at, batch, File and directory operations, file attributes Working with zip files and long files.

Activity: System Monitor Script.

Textbook 1: 7,8,5,4,6,15

Module-3 User Management	**Hours: 8**

Creating, modifying, and deleting users and groups, Account management and permissions.

Activity: Batch User Creation and Permission Management

Textbook 1:12,13,14,15,16

Module-4 Introduction to Git	**Hours: 8**

Git installation and initialization, Staging and committing files, Branching and merging, Stashing and applying changes, Cloning, pushing, pulling from remote repositories.

Activity: Create and Manage Git Repository for Personal Notes

Textbook 2: chapter 1.5-1.8,2,3	
Module-5 Git Workflows and Advanced Git Operations	**Hours: 8**
Creating tags and release management, Viewing specific commits and log filtering, Analysing commit history by author and date. **Activity:** Team Collaboration Simulation with Git Textbook 2: 6,7.1-7.3,8,9	

Suggested Learning Resources: **Textbooks:**	
1	Yashavant P. Kanetkar, Unix Shell Programming, 1st edition, BPB Publications, 2003.
2	Scott Chacon and Ben Straub, Pro Git, 2nd Edition, Published by Apress, 2014.
Reference Books:	
1	Michael Kerrisk, The Linux Programming Interface: A Linux and UNIX System Programming Handbook, 1st edition, Reprinted, No Starch Press 2023.
2	Cameron Newham, Learning the Bash Shell, 3rd Edition, O'Reilly Media, 2005.
3	Richard E. Silverman, Git Pocket Guide, 1st edition, O'Reilly Media, 2013.
4	Richard Blum and Christine Bresnahan, Linux Command Line and Shell Scripting Bible, 4th Edition, Wiley, 2021.

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning

- Linux Admin Automation with Shell Script.
- Collaborative Project with Git Workflow.

Course outcomes:	
CO1	Identify and apply UX design principles to create intentional and user-centered designs.
CO2	Analyze and demonstrate the complete UX design lifecycle, including processes, tools, and outcomes.
CO3	Utilize industry-standard UX tools to prepare professional project deliverables.
CO4	Design decisions through effective demonstration and assessment of UX artifacts.
CO5	Use Git workflows, tagging, and history manipulation for effective version control.

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	2	-	-	-	-	-	-	-
CO2	3	3	2	-	2	-	-	-	-	-	-	-
CO3	3	2	2	-	2	-	-	-	-	-	-	-
CO4	2	2	3	-	3	-	-	-	-	-	-	-
CO5	2	3	2	2	3	-	-	-	-	-	-	-

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Operating System	**Semester**	VII
Course Code	CS432	**CIE**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week L:T:P	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Dept	CS		

PREREQUISITES: C Programming Course.

Course objective is to:

1. Introduce the concepts, roles, and functions of traditional and modern operating systems.
2. Explain the concept of process and its management which includes process scheduling.
3. Understand the problems related to concurrency, different synchronization mechanisms and deadlock handling.
4. Illustrate the various memory management and file management techniques.
5. Understand different APIs/Commands related to processor, memory, storage, and filesystem management.

Module-1: Overview and structure of Operating System — **8 Hours**

Overview Operating Systems: Definition - Operating System, computing environment and nature of computations, classes of operating systems, Efficiency, system performance and user services, Batch processing systems, multiprogramming systems, time sharing systems, real time OS, distributed OS, Modern OS.

Structure of Operating Systems: operation of an OS, structure of an OS, layered design of OS, Virtual machine OS.

Text book 1: Chapter 3.1 to 3.9, 4.1 to 4.4

Module-2: Process, Threads and Process Synchronization — **8 Hours**

Process and Threads: Processes and programs, Implementing process, case study: Threads, processes in UNIX.

Process Synchronization: Race conditions, Critical section, control synchronization and invisible operations, synchronization approaches, structure of concurrent systems, classic approach synchronization problems, semaphores, monitors, case study: process synchronization in UNIX.

Textbook 1: Chapter 5.1 to 5.3, 5.4.1, 6.1 to 6.7, 6.9, 6.10, 6.11.2.

Module-3: Scheduling and Deadlocks — **8 Hours**

Scheduling: Scheduling Terminology and concepts, non-pre-emptive and pre-emptive scheduling policies, scheduling in practice, Case study: scheduling in UNIX.

Deadlocks: what is deadlock? Deadlocks in resource allocation, Handling Deadlocks, Deadlock Detection and Resolution, Deadlock prevention, deadlock avoidance, case study: Deadlock handling in UNIX.

Textbook 1: Chapter 7.1 to 7.4, 7.6.1, 8.1 to 8.6, 8.8.1.

Module-4: Memory Management and Virtual memory — **8 Hours**

Memory Management: Managing the memory hierarchy, static and dynamic allocation, execution of program, memory allocation to process, contiguous and non-contiguous memory allocation, paging, segmentation, segmentation with paging.

Virtual memory: Basics, demand paging, the virtual memory manager, page replacement policies, Case study: UNIX virtual memory

Textbook 1: Chapter 11.1 to 11.4, 11.6 to 11.10

Module-5: File Systems and Mass Storage Structures	**8 Hours**

File Systems: overview of File Processing. Files and file operation, fundamental file organization and access methods, directories, case study: Unix File system.

Mass Storage Structures: Overview; Disk Structure, Scheduling, Management.

Textbook 1: Chapter 13.1 to 13.4, 13.14.1 Textbook 2: chapter 12.1,12.2,12.4,12.5.

Follow-on Courses: System Software Compiler Design, Embedded Systems and Cloud Computing.

Resources for Advanced Learners

An Introduction to Operating Systems Concepts and Practice, PCP Bhatt, 2013 https://www.amazon.in/INTRODUCTION-OPERATING-SYSTEMS-CONCEPTS-PRACTICE-dp-9387472876/dp/9387472876/ref=dp_ob_title_bk

Open ended experiments leading to guided projects:

1. Develop a lightweight Operating System

Suggested Learning Resources:

Textbooks:

1.	Dhananjay M Dhamdhere, Operating systems: a Concept- Based Approach, 3rd Edition, McGraw hill, 2012.
2.	Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, Operating System Concepts, Inc., 9th Edition, John Wiley & Sons ,2012.
3.	Unix Concepts and Applications, 4th Edition, Sumitabha Das, TMH, 2017.

Reference Books:

1.	Ann McHoes Ida M Fylnn, Understanding Operating System, 6th Edition, Cengage Learning ,2010.
2.	William Stallings, “Operating Systems: Internals and Design Principles”, Eighth Edition, Prentice Hall, 2015.
3.	Unix Shell Programming, Yashwant Kanetkar, 1st Edition, BPB Publications, 2003.
4.	Introduction to UNIX by M G Venkatesh Murthy, 1st Edition, Pearson Education, 2009.
5.	Practical Linux System Administration. by Kenneth Hess, 1st Edition, O'Reilly Media, Inc, 2023.
6.	Beginning Linux System Administration. by Sander van Vugt, 1st Edition, Pearson Education, 2016.

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning

1. Setting up a VMware Workstation
2. Installing a VMware Guest OS
3. Running LINUX Operating System

Course outcomes:	
CO1	Identify the roles and functions of traditional and modern operating systems.
CO2	Analyse the various techniques for process management and process scheduling
CO3	Evaluate the problems related to concurrency, different synchronization mechanisms and deadlock handling.
CO4	Apply various technique for memory management
CO5	Analyse file and secondary storage management strategies

CO-PO Mapping

CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	-	-	-	-	-	-	-	-	2
CO2	3	2	3	-	-	-	-	-	-	-	-	2
CO3	3	2	3	-	2	-	-	-	-	-	-	2
CO4	3	2	3	1	2	-	-	-	1	1	-	2
CO5	3	2	3	-	-	-	-	-	-	-	-	2

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Introduction to Blockchain	**Semester**	VII
Course Code	CS433	**CIE**	50
Total No. of Contact Hours	40	**SEE**	50
No. of Contact Hours/week (L:T:P)	3:0:0	**Total**	100
Credits	3	**Exam. Duration**	3 Hours
Teaching Department	CS		

Course Overview

This course provides an in-depth understanding of distributed computing and blockchain technology, focusing on their fundamental principles, architecture, and real-world applications. It explores how blockchain enables decentralized computing and secure digital transactions, with an emphasis on Bitcoin and cryptocurrencies.

Course objective is to:

1. Explain the fundamentals of distributed computing and blockchain.
2. To explore the driving force behind the cryptocurrency Bitcoin, along with the Decentralization and Cryptography.
3. Familiarize with Bitcoins and its alternative coins, Smart contracts and outside of currencies.

Prerequisite Knowledge: A Basic knowledge in Data structures, Cryptography, Networking & Programming Languages

Module-1	**8 Hours**

Blockchain Concepts: Distributed systems, History of blockchain, Introduction to blockchain, Types of blockchain, CAP theorem and blockchain, Benefits and limitations of blockchain.

Activity: Case Study Discussion, Debate, MCQs

Textbook 1: Chapter 1

Module-2	**8 Hours**

Decentralization and Cryptography: Decentralization using blockchain, Methods of decentralization, Routes to decentralization, Decentralized organizations. **Introduction to Cryptography & Cryptocurrencies:** Cryptographic Hash Functions, Hash Pointers and Data Structures, Digital Signatures, Public Keys as Identities, A Simple Cryptocurrency.

Activity: Hash Function Challenge-Students manually calculate simple hash values for provided inputs , Bulid cryptocurrency system:Groups create a mock cryptocurrency system using paper tokens. Each group maintains a ledger of transactions and uses "proof-of-work" (e.g., solving a math puzzle) to validate transactions.

Textbook 1: Chapter 2, Textbook 2: Chapter 1

Module-3	**8 Hours**

Bitcoin and Alternative Coins A: Bitcoin, Transactions, Blockchain, **Bitcoin payments B**: Alternative Coins, Theoretical foundations, Bitcoin limitations, Namecoin, Litecoin, Primecoin, Zcash

Activity: Comparison Table to understand differences and similarities between Bitcoin and alternative coins. Group Presentation: Solving Bitcoin Limitations

Textbook 1: Chapter 4, 5

Module-4	**8 Hours**

Smart Contracts and Ethereum: Smart Contracts: Definition, Ricardian contracts. **Ethereum:** Introduction, Ethereum blockchain, Elements of the Ethereum blockchain, Precompiled contracts.

Activity: Ethereum Blockchain Visual Map:Provide students with blank diagrams of the Ethereum blockchain structure & make them fill, Quiz

Textbook 1: Chapter 6, 7.1 to 7.4

Module-5	**8 Hours**

Alternative Blockchains: Blockchains **Blockchain-Outside of Currencies:** Internet of Things, Government, Health, Finance, Media.

Activity: Industry Use Case Matching:Provide cards with industry names and matching use cases.

Textbook 1: Chapter 10.1,11

Follow on courses: Advanced Cryptography & Security, Decentralized Finance (DeFi) & Smart Contracts, Blockchain Scalability & Consensus Mechanisms

Suggested Learning Resources:
Textbooks:

1.	Mastering Blockchain - Distributed ledgers, decentralization and smart contracts explained, Second Edition, Imran Bashir, Packt Publishing Ltd, 2017.
2.	Arvind Narayanan, Joseph Bonneau, Edward W. Felten, Andrew Miller, Steven Goldfeder and Jeremy Clark., Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction, 1st edition, Princeton University Press, 2016.

Reference Books:

1.	Bitcoin and Cryptocurrency Technologies, Author- Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder, 1st edition, Princeton University, 2016.
2.	Blockchain Basics: A Non-Technical Introduction in 25 Steps, Author- Daniel Drescher, First Edition, A press ,2017.
3.	Mastering Bitcoin: Unlocking Digital Cryptocurrencies, Andreas M. Antonopoulos, First Edition, O'Reilly Media, 2014.

Web links and Video Lectures (e-Resources):

M1: https://nptel.ac.in/courses/106/105/106105184/
M2: https://ethereum.org/en/developers/

Activity-Based Learning (Suggested Activities in Class)/ Practical Based learning
Quiz, Group Discussions, Seminars, In class assignments.

Course outcomes:	
CO1	Outline the types, benefits and limitation of blockchain.
CO2	Outline the blockchain decentralization and cryptography concepts.
CO3	Assess the Bitcoin features and its alternative options.
CO4	Outline the smart contracts
CO5	Identify the blockchain features outside of currencies.

CO-PO Mapping

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	3	2	-	-	-	3	1	-	-	-	-	2
CO2	3	2	2	-	-	1	2	-	-	-	-	2
CO3	3	2	2	-	-	3	1	-	-	-	-	2
CO4	3	2	2	-	-	2	2	-	-	-	-	2
CO5	3	2	-	-	-	-	-	-	-	-	-	2

High-3, Medium-2, Low-1

Note: All modules in this course are designed to address Learning Levels L1 to L4

Course Title	Internship	**Semester**	VIII
Course Code	CS451	**CIA**	50
Total No. of Contact Hours	--	**SEE**	50
No. of Contact Hours/week (L:T:P)	---	**Total**	100
Credits	10	**Exam. Duration**	---
Teaching Dept	CS		

A Research Internship provides students with the opportunity to gain hands-on experience in academic or industrial research under the mentorship of a faculty member or research professional. During the internship, students engage in activities such as literature review, data collection and analysis, experimental design, model development, and technical documentation. The focus is on developing a deeper understanding of research methodologies, critical thinking, and problem-solving skills in a specialized domain. The internship culminates in a final report and/or presentation, demonstrating the student's contribution to the research and their ability to apply academic knowledge in a practical research setting.

Indicative List of UG 8th Semester Online Courses

As part of the **Professional Elective Course** – **CS452** in the eighth semester, students are required to complete an **online certification course**. Students may choose a 3-credit online course either from NPTEL (Swayam); however, they must appear for the proctored examination and submit the certificate upon successful completion.

List of NPTEL Courses / Industry Certification Courses

Sl.No	Course ID	Course name	Course Link
1	noc25-cs87	Foundations of Virtual Reality	https://onlinecourses.nptel.ac.in/noc25_cs91/preview
2	noc25-cs93	Deep Learning for Computer Vision	https://onlinecourses.nptel.ac.in/noc25_cs93/preview
3	noc25-cs96	Parameterized Algorithms	https://onlinecourses.nptel.ac.in/noc25_cs96/preview
4	noc25-cs97	Getting Started with Competitive Programming	https://onlinecourses.nptel.ac.in/noc25_cs97/preview
5	noc25-cs102	Social Networks	https://onlinecourses.nptel.ac.in/noc25_cs102/preview
6	noc25-cs113	Software Testing	https://onlinecourses.nptel.ac.in/noc25_cs113/preview
7	noc25-cs117	Privacy and Security in Online Social Media	https://onlinecourses.nptel.ac.in/noc25_cs117/preview
8	noc25-cs118	Responsible & Safe AI Systems	https://onlinecourses.nptel.ac.in/noc25_cs118/preview
9	noc25-cs120	Practical Cyber Security for Cyber Security Practitioners	https://onlinecourses.nptel.ac.in/noc25_cs120/preview
10	noc25-cs135	Design & Implementation of Human-Computer Interfaces	https://onlinecourses.nptel.ac.in/noc25_cs135/preview
11	noc25-cs144	Programming in Modern C++	https://onlinecourses.nptel.ac.in/noc25_cs144/preview
12	noc25-cs146	Introduction to Industry 4.0 and Industrial Internet of Things	https://onlinecourses.nptel.ac.in/noc25_cs146/preview
13	noc25-cs161	Introduction to Large Language Models (LLMs)	https://onlinecourses.nptel.ac.in/noc25_cs161/preview

***** As per the scheme for the 2023–24 batch, a 3-credit online course is part of the curriculum for 8th semester students. The selected courses are offered through NPTEL(swayam), consisting of video lectures and culminating in a proctored exam. The score reflected in certificate shall be directly used for grade processing.

Note: Students must ensure that any professional elective they choose does not have the same title or substantially similar content as any other course completed during their entire 4-year degree program.

Indicative List of UG 8th Semester Online Courses

As part of the **Open Elective Course – YY453** in the eighth semester, students are required to complete an **online certification course**. Students may choose a 3-credit online course either from NPTEL (Swayam); however, they must appear for the proctored examination and submit the certificate upon successful completion.

List of NPTEL Courses / Industry Certification Courses

Sl.No	Course ID	Course name	Course Link
1	noc25-cs117	Privacy and Security in Online Social Media	https://onlinecourses.nptel.ac.in/noc25_cs117/preview
2	noc25-cs116	Cyber Security and Privacy	https://onlinecourses.nptel.ac.in/noc25_cs116/preview
3	noc25-cs146	Introduction to Industry 4.0 and Industrial Internet of Things	https://onlinecourses.nptel.ac.in/noc25_cs146/preview
4	noc25-ag15	Operations Research	https://onlinecourses.nptel.ac.in/noc25_ag15/preview

***** As per the scheme for the 2023–24 batch, a 3-credit online course is part of the curriculum for 8th semester students. The selected courses are offered through NPTEL(swayam), consisting of video lectures and culminating in a proctored exam. The score reflected in certificate shall be directly used for grade processing.

Note: Students must ensure that any professional elective they choose does not have the same title or substantially similar content as any other course completed during their entire 4-year degree program.